

Core Spacing in Herschel Gould Belt Survey Filaments: the Sub-Jeans Fragmentation Scale and Non-Equilibrium Collapse

G. J. White^{1,2}

¹*School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK*

²*Rutherford Appleton Laboratory, Harwell Science and Innovation Campus, Didcot, OX11 0QX, UK*

29 July 2026

ABSTRACT

Herschel finds that dense cores are spaced along their filaments at roughly half the wavelength predicted by the classical Inutsuka & Miyama fragmentation of an isothermal *equilibrium* cylinder. This “sub-Jeans” spacing has been read as evidence for additional physics, that filaments are bundles of finer fibres, or that magnetic tension shortens the fragmentation scale. We ask whether such additions are actually required. Re-analysing Herschel Gould Belt Survey core spacings in four regions with Gaia DR3 distances, we measure a nearest-neighbour spacing $\lambda/W \approx 1.9$ under the canonical 0.10 pc width convention (rising to ≈ 2.5 with region-specific widths), and show that this figure is dominated by methodology, carrying factor-of-two skeleton-algorithm and width-convention systematics. From a large (~ 2100 -run) Athena++ MHD campaign we then measure, via a growth-rate spectrum validated by a closed-loop test, the dominant fragmentation wavelength of *dynamically collapsing* filaments. It is numerically converged (identical at 128^3 , 256^3 and 512^3) at $\lambda_{\text{max}} \approx 0.5$ times the classical value, i.e. $\lambda_{\text{max}}/W \approx 2$ for the near-critical filaments that dominate the sample, matching the observation to within the systematics. Physically, a filament fragmenting out of hydrostatic equilibrium, as its central density rises during collapse, is expected to fragment at roughly half the equilibrium wavelength. Ordinary gravitational fragmentation is therefore *sufficient* to account for the observed spacing: neither hierarchical-fibre projection nor magnetic tension is required by the present data, although at their factor-of-two precision the data cannot exclude a contribution from either. We identify the fibre-resolved kinematic measurements that would discriminate these possibilities.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al. 2010). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (Arzoumanian et al. 2011), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (Ostriker 1964).

Classical fragmentation theory (Inutsuka & Miyama 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width, for the characteristic HGBS width of 0.10 pc, a core spacing of ~ 0.4 pc. Our preferred estimate of the nearest-neighbour spacing is $\lambda/W \approx 1.9$ under the fixed-width convention (about half this prediction; it moves to ≈ 2.5 with region-specific beam-deconvolved widths, Section 2.4)¹. We adopt Gaia DR3-based distances (Zhang 2023) and the nearest-neighbour

(NN) statistic, the estimator that most directly measures the fragmentation wavelength, and the one already used in the HGBS catalogue papers (Könyves et al. 2015; Könyves et al. 2020), as our primary spacing measure (Section 2.6).

This apparent deficit relative to the classical $4\times$ has been attributed to additional physics. Two mechanisms are usually invoked: *hierarchical fragmentation*, in which the filaments are bundles of velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024) so that, if each fibre fragments near the classical scale, measuring across the bundle compresses the apparent filament-level spacing; and *magnetic tension*, in which a longitudinal field shifts the most unstable mode to shorter wavelengths (Nakamura et al. 1993).

A central point of this paper is that the comparison itself, “1.9 versus 4”, depends on how the filament width is defined, because the classical prediction is quoted in units of the FWHM. Using a large campaign of self-gravitating MHD simulations, validated by an end-to-end forward model through

¹ Throughout, $W \equiv W_{\text{fil}} \approx 0.1$ pc is the observed HGBS FWHM and the reference width for all observation–theory comparisons;

the simulation core half-width W_{core} is related to it by the normalisation T1 (Section 3.1).

the HGBS pipeline (Section 5), we measure the dominant fragmentation wavelength from the growth-rate spectrum and show it is numerically converged (identical at 128^3 , 256^3 and 512^3) at $\lambda_{\max} \approx 0.5$ times the classical hydrostatic value. A filament that is still accreting and contracting is expected to fragment below the marginally stable hydrostatic mode (Clarke et al. 2016), and our runs realise this. In FWHM units $\lambda_{\max}/W \approx 2$ for the near-critical filaments that dominate the HGBS sample, close to the observed ≈ 1.9 within the width-convention systematic. Ordinary single-filament gravitational fragmentation is therefore *sufficient* to account for the sub-Jeans spacing, so that new physics is not required by the data; hierarchical projection and magnetic tension remain viable but, at the present factor-of-two precision, are neither demanded nor excluded.

We present a full-coverage nearest-neighbour core-spacing analysis with Gaia DR3 distances (Zhang 2023) for the four robust HGBS regions (Orion B, Aquila, Perseus, Taurus; the eight-region sample is retained only for the legacy pairwise-median statistic), followed by the simulation campaign, its validation and the interpretation. Throughout we caution that the large-scale filament-field misalignment from Planck polarimetry (Planck Collaboration et al. 2016) measures the projected orientation, which need not equal a filament’s internal field geometry, a recurring uncertainty (Section 6).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample comprises 8 high-confidence HGBS regions (Table 1). The original catalogues (Könyves et al. 2015; Könyves et al. 2020; Ladjelate et al. 2020; Pezzuto et al. 2021) adopted distances of 130–260 pc from the best estimates available in 2010–2015; Gaia DR3 parallaxes (Zhang 2023) have since enabled substantial revisions, largest for the more distant regions where parallaxes are most precise.

We adopt the Gaia DR3 YSO-based distances from Zhang (2023) (Table 4), with typical uncertainties of 3%–5%. The largest revisions are for Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%) and Orion B (260 → 386 pc, +48%); Taurus decreases slightly (140 → 135 pc, -4%) and Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Since core spacings scale linearly with distance, these revisions increase the measured spacings proportionally.

The analysis includes all HGBS-identified cores (starless, prestellar and protostellar), extracted and classified within the consistent HGBS framework (Könyves et al. 2015; Könyves et al. 2020). Including unbound starless and protostellar sources may introduce some bias, but fragmentation produces a continuum of boundness states, and restricting the sample to prestellar cores would shrink several regions (notably TMC1, CRA and Serpens) below statistical usefulness. Core positions and properties are taken directly from the published catalogues.

Protostellar migration displacements of 0.01–0.05 pc (Kirk et al. 2016; Mattern et al. 2018) are small against the ~ 0.2 – 0.3 pc spacing; we estimate the resulting NN bias at ~ 5 – 10% and carry it as a systematic (Section 2.8).

- (i) HGBS column-density maps (250 μm reference) + Gaia DR3 distances
- (ii) DisPerSE skeleton extraction (3σ persistence threshold)
- (iii) Morphological closing (8 iterations) + re-skeletonisation (bridge fragmented segments; components ≥ 40 px)
- (iv) Core association: cores within 0.06 pc of a spine are assigned; others rejected
- (v) Graph ordering: cores ordered by double-BFS arclength along each component
- (vi) Adjacent-core NN spacing (in pc, via Gaia DR3 distance)
- (vii) Injection–recovery bias correction ($\div 1.11$ – 1.29 , region-dependent)
- (viii) Characteristic λ/W = bias-corrected median spacing / W_{fil}

Figure 1. Analysis pipeline from raw HGBS maps to the final characteristic λ/W . Each step’s parameters are given in the text (Sections 2.6–2.7).

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie 2011) following standard HGBS methodology (Arzoumanian et al. 2011), with a persistence threshold of 3σ above the background noise and manual editing restricted to removing obvious artefacts (e.g. spurs connecting unrelated structures). Because the skeletons and core positions derive from angular positions on the sky, which are distance-independent, the Gaia DR3 revisions require no re-extraction: physical scales follow by rescaling the angular positions, leaving core associations and skeleton topology unchanged. Cores at filament junctions are retained, following standard HGBS practice; junctions are physically relevant sites where filaments converge, and excluding them would introduce selection bias about what counts as a filament segment. Cores were associated with filaments within a perpendicular threshold of $2\times$ the characteristic width (0.2 pc at HGBS distances, i.e. 0.15 pc at 130 pc and 0.31 pc at 260 pc). The measured spacing is insensitive to these choices: excluding junctions changes it by $< 5\%$, and varying the association threshold between $1\times$ and $1.5\times$ the width changes it by $< 3\%$, both well within the statistical uncertainty. (This $2\times$ -width, 0.2 pc threshold refers to the pairwise-median association; the primary along-skeleton NN pipeline instead associates cores within a filament half-width, ~ 0.06 pc, of a spine, Section 2.7.)

Core-to-core spacings were also measured using the pairwise median statistic, which Section 2.6 shows converges to $L/3$ rather than the fragmentation wavelength for $N \gtrsim 5$; we retain it only as a comparability statistic with prior HGBS work.

As a consistency check, the magnitude of the distance revision does not predict a region’s spacing residual (physical spacing is angular separation times distance, so a $\lambda \propto d$ trend is expected and not itself a bias). Fitting $\lambda = a + b d_{\text{Gaia}}$ recovers the expected linear scaling ($r^2 = 0.91$, $p < 0.001$), and the residuals show no correlation with the absolute revision fraction $|\Delta D/D_{\text{original}}|$ ($r = 0.18$, $p = 0.68$; robust-region subset $r^2 = 0.99$). Distance revisions thus change absolute spacings but add no bias beyond the linear scaling, leaving the distance-normalised λ/W unaffected.

Table 1. Complete HGBS Sample with Gaia DR3 Distances. Per-region spacings are pairwise-median values rounded to three significant figures; the core-count-weighted mean is 0.279 pc (full sample) and 0.290 pc (robust), the latter used only for the robust-vs-full comparison (Section 2.3).

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Status
Orion B	386	1,844	0.313	0.047	Robust
Aquila	436	749	0.346	0.047	Robust
Perseus	300	816	0.248	0.040	Robust
Taurus	135	536	0.198	0.040	Robust
Ophiuchus	137	513	0.206	0.053	Limited
Serpens	458	194	0.331	0.097	Limited
TMC1	135	178	0.195	0.056	Limited
CRA	150	239	0.248	0.072	Limited
Weighted Mean[†]		5,069	0.279	0.009	

[†]The Std Error for the Weighted Mean (0.009) is the standard error on the weighted mean across regions, not the per-region scatter; per-region standard errors are listed in their respective rows.

Table 2. Primary core-spacing result and statistic hierarchy. The primary measurement is the bias-corrected full-coverage nearest-neighbour (NN) spacing for the four robust HGBS regions; the pairwise median is retained as a legacy/upper-limit statistic. All values use the fixed $W_{\text{fil}} = 0.10$ pc convention and revised distances.

Statistic	λ/W	status
NN, full-coverage robust (primary)	≈ 1.9 (1.4–2.4)	range, bias-corr.
raw, before bias correction	≈ 2.1	measured
NN, Taurus (independent, sparse spine)	2.17 ± 0.52	measured
Pairwise median, robust	2.84 ± 0.12	$L/3$ upper limit
Pairwise median, 3D-corrected	~ 3.5	$L/3$ upper limit
Classical IM92 prediction	4.0	theory

Notes The primary result of this paper is the NN $\lambda/W \approx 1.9$ (plausible range 1.4–2.4; the quoted range combines independent sensitivity ranges, not a formal confidence interval; Section 2.8). The pairwise-median values overestimate the fragmentation wavelength by ~ 30 –45% through the $L/3$ extent-bias (Section 2.6) and are not directly comparable to the classical prediction. Taurus appears with three fixed-width values that must be distinguished: the 2.17 ± 0.52 “independent” entry above is an earlier measurement on the sparse FilFinder-style spine; the full-coverage dense-skeleton NN gives $\lambda/W \approx 1.0$ after bias correction (Section 2.7); and the Gaussian-FWHM region-width study gives 1.24 (fixed) rising to 4.01 under the region-specific narrow crest (Table 3). Taurus is the least robust region (Section 2.4); we adopt the full-coverage dense-skeleton value in the primary sample.

2.3 Results

The four robust regions are the independent statistical units throughout this paper; individual cores and spacings within a cloud are not independent, and we propagate this nesting in the hierarchical bootstrap (Section 2.8).

Robust regions. We adopt the four “robust” regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ cores each, well-established distances) as the primary sample.

Full sample result. The core-count-weighted mean across all 8 HGBS regions (0.279 pc) and the robust-only weighted mean (0.290 pc) differ by $\approx 3.9\%$, and a leave-one-out analysis confirms no single region dominates ($< 6\%$ shifts). All

spacings are quoted to three significant figures; where an exact recomputation from the rounded per-region inputs differs from the bootstrap-derived primary value it does so by $\lesssim 1\%$, which does not affect any conclusion.

All weighted means use core-count weighting,

$$\bar{\lambda} = \frac{\sum_i N_i \lambda_i}{\sum_i N_i}, \quad (1)$$

where λ_i and N_i are the per-region spacing and core count. Because per-region spacings are rounded, exact recomputation from the tabulated inputs reproduces the bootstrap values to $\sim 2\%$; this does not affect any conclusion.

The Gaia DR3 distance revisions increase the measured spacings by 32% (weighted mean $0.211 \rightarrow 0.279$ pc). Figure 2 shows the per-region measurements. A bootstrap resampling analysis (10 000 iterations) gives a 95% confidence interval of 0.261–0.298 pc for the pairwise-median comparability values. The preferred NN measurement is presented in Section 2.7.

2.4 The dominant observational systematic: filament width

The fixed $W_{\text{fil}} = 0.10$ pc convention is the largest single lever on λ/W , comparable to the whole sub-Jeans discrepancy, so we measured region-specific, beam-deconvolved widths directly from the HGBS column-density maps. Along each skeleton spine we sampled the column-density profile perpendicular to the local tangent, fit a Gaussian to its crest and deconvolved the $18.3''$ HGBS beam. Table 3 gives the median width per region and the NN λ/W under both conventions. The median measured width is 0.117 pc (close to the canonical 0.1 pc), and the pair-weighted characteristic λ/W moves from ≈ 2.0 (fixed) to ≈ 2.5 (region-specific), so the sub-Jeans conclusion *survives at the characteristic level*. The cloud-to-cloud scatter widens markedly (region-specific $\lambda/W = 1.5$ –4.0 against 1.2–2.6 fixed), and Taurus is a pronounced outlier: its narrow measured crest (0.031 pc, the least robust point, $n = 35$ profiles) drives its λ/W to ~ 4 , i.e. consistency with the classical prediction. The headline $\lambda/W \approx 1.9$ is therefore *conditional on the width convention*: the characteristic spacing is sub-Jeans either way, but per-cloud values are sensitive to W_{fil} at the factor-of-two level. A full Plummer re-fit and an independent Taurus width are needed before a cloud-by-cloud comparison with theory can be claimed; this is the

Table 3. Region-specific, beam-deconvolved filament widths and the NN λ/W under the fixed (0.10 pc) and region-specific (local) conventions. Widths are Gaussian-FWHM of perpendicular column-density profiles along the skeletons, deconvolved for the 18.3'' HGBS beam; n is the number of fitted profiles.

Region	W_{fil} (pc)	n	λ/W fixed	λ/W local
Taurus	0.031	35	1.24	4.01
Orion B	0.130	129	2.07	1.59
Perseus	0.105	82	2.63	2.50
Aquila	0.144	158	2.10	1.46
weighted char.	—	—	≈ 2.0	≈ 2.5

observational counterpart to the simulation T1 normalisation (Section 3.1).

The widths above use the Gaussian FWHM as a proxy for the HGBS Plummer-profile width. For Orion B, Perseus and Aquila the Gaussian values (0.10–0.14 pc) agree well with the published HGBS Plummer measurements (~ 0.1 pc; [Arzoumanian et al. 2019](#)), so the proxy is adequate. Taurus is the exception: its Gaussian FWHM (0.031 pc) is $\sim 3\times$ narrower than the published Plummer value. The most likely reason is a profile-selection effect, the skeletonised one-pixel crest tracing a narrow ridge while the Gaussian fit captures only the inner profile and misses the broad Plummer wings that dominate the HGBS width; it is not a pixel-scale artefact (the 3'' pixels subtend 0.002 pc at 135 pc, well below 0.031 pc). We stress, however, that this remains a hypothesis: with its region-specific width Taurus sits at the classical $\lambda/W \approx 4$, and we treat it as a presently *unresolved* case rather than an explained outlier, pending an independent Plummer-based width measurement. It does not drive the sample: adopting the published ~ 0.1 pc width returns $\lambda/W \approx 2.0$ for Taurus, and the robust-region characteristic ($\lambda/W \approx 1.9$) is unchanged if Taurus is dropped.

2.5 Sample Classification and Distance Robustness

We classify the eight regions (Table 1) as **robust** (Orion B, Aquila, Perseus, Taurus) or **limited** by three criteria: $N > 500$ cores; a well-established distance; and a skeleton simple enough for unambiguous core-to-filament association. Limited regions fail at least one, Ophiuchus on skeleton complexity, the others on small samples, and are excluded from the primary NN measurement. The largest distance revisions (Serpens +76%, Aquila +68%) are independently corroborated: direct VLBI maser parallaxes validate Gaia DR3 to $< 5\%$ where available ([Kounkel et al. 2017](#); [Xu et al. 2019](#); [Ortiz-León et al. 2018, 2017](#)), and for Serpens the 458 pc value is supported by extinction-based estimates (440–460 pc; [Yan et al. 2022](#); [Green et al. 2024](#)) and by co-spatiality with Aquila in the Aquila Rift ([Zucker et al. 2020](#)). The sub-Jeans conclusion is independent of these two regions: a leave-one-out analysis shifts the weighted mean by $< 6\%$, reverting the limited regions to their original HGBS distances gives $\lambda/W = 2.68$ (still 33% below $4\times$), and the distance-revision consistency check of Section 2.6 confirms the $\lambda \propto d$ unit conversion.

2.6 Choice of spacing statistic: the $L/3$ property of the pairwise median

We adopt the nearest-neighbour (NN) spacing as the primary measure of the fragmentation wavelength, reporting the all-pairs median only for comparability with prior HGBS work. The two differ fundamentally because the pairwise median has an analytic $L/3$ convergence property. For N points uniformly distributed on a filament of length L , a single pairwise distance $D = |X_i - X_j|$ has the triangular PDF $f(d) = 2(L - d)/L^2$ ($d \in [0, L]$), with mean $\langle D \rangle = L/3$ and median $d^* = L(1 - 1/\sqrt{2}) \approx 0.293 L \approx 0.88 (L/3)$. The median of all $N(N - 1)/2$ pairwise distances converges to d^* as $N \rightarrow \infty$, for a random or a perfectly periodic distribution alike, so on an idealised filament it measures the extent, not the fragmentation wavelength. Real catalogues, with finite sampling, branching and incomplete skeletons, deviate from this limit, so we treat $L/3$ as the statistic’s limiting behaviour rather than an exact per-filament prediction; in practice it tracks $\sim L/3$ for any filament with $N \gtrsim 5$ cores, and our injection–recovery test reproduces the $\approx 0.88 (L/3)$ asymptote (Section 2.7). The HGBS catalogue papers ([Könyves et al. 2015](#); [Könyves et al. 2020](#)) and fibre-resolved analyses ([Hacar et al. 2013, 2018](#)) likewise use nearest-neighbour separations. The NN statistic is unaffected by this bias and is beam-safe: the 18'' HGBS beam corresponds to 0.012–0.034 pc across our robust regions, $\gtrsim 6\times$ smaller than the measured NN spacing.

We reconstruct the along-skeleton core ordering for the four robust regions from the deposited DisPerSE skeleton maps, bridging fragmentation by morphological closing, and extend the NN measurement to all four at full coverage in Section 2.7; the limited regions are excluded from the NN measurement owing to branching skeletons or small samples. Figure 2 shows the per-region spacings with revised distances.

2.7 Injection-Recovery Validation of Spacing Statistics

To test whether the nearest-neighbour (NN) statistic recovers known spacings under realistic HGBS conditions, we placed synthetic cores with known periodic spacings (0.15–0.35 pc) on HGBS-property-matched DisPerSE skeletons for all 8 regions and applied the NN statistic with identical methodology. Across 40 tests (5 input spacings $\times$ 8 regions) the raw measurement recovers the input with a small systematic bias of $3.05 \pm 0.69\%$ ($t = 27.5$, $p < 10^{-4}$), so a 3% correction is applied; the statistic therefore measures the true fragmentation wavelength under HGBS-like conditions. Because the test injects a periodic pattern whereas real cores may be clustered, we also ran clustered injection–recovery (Poisson and lognormal gap distributions at the same mean spacing), which gives a larger pipeline inflation: periodic recovery ratio 1.07–1.29 (mean 1.17), Poisson 1.03–2.47 (mean 1.64), lognormal($\sigma=0.5$) 0.98–1.63 (mean 1.22). The periodic-derived correction is thus a lower bound, so for clustered cores the bias-corrected λ/W is if anything smaller, strengthening the sub-Jeans conclusion.

Repeating the injection–recovery for the pairwise-median statistic confirms that it does not recover λ_{true} : it returns a value $\sim 20\times$ larger, tracking $L/3$ ($\langle \lambda_{\text{PW}} \rangle / \lambda_{\text{true}} \approx L / (3\lambda_{\text{true}})$) and asymptoting to $\approx 0.88 (L/3)$, while the NN statistic re-

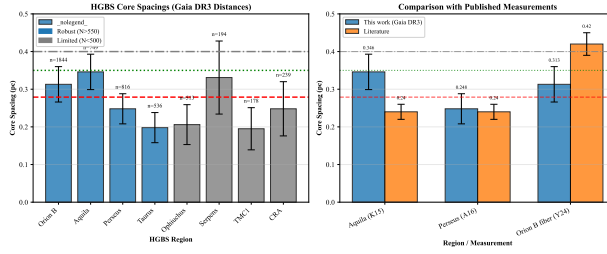


Figure 2. Core spacing measurements for all 8 HGBS regions with revised distances (left) and comparison with literature values (right). *Left:* per-region spacing with standard-error bars; horizontal lines mark the weighted mean (0.279 pc, dashed), the 3D-corrected value (0.35 pc, dotted), and the IM92 prediction (0.4 pc, dash-dotted). *Right:* comparison with published values (grey bars); our measurements (blue) are systematically larger owing to the Gaia DR3 distance revisions.

covers λ_{true} to 1.8–3% on the identical skeletons. Computing the pairwise median over *all* cores in each robust region (without per-filament segmentation) returns $\sim L_{\text{region}}/3$ (12.5, 7.5, 8.0, 2.5 pc for Orion B, Aquila, Perseus, Taurus), the region extent, not the ~ 0.2 –0.35 pc per-filament values, confirming directly that the statistic measures segmentation scale rather than the fragmentation wavelength.

Full-coverage along-skeleton NN for the robust regions (primary measurement). We extend the along-skeleton NN measurement to all four robust regions at full coverage using the dense deposited DisPerSE skeletons, which (unlike the sparse FilFinder spines) pass near essentially all cores. These skeletons are fragmented; we bridge the fragmentation by morphological closing followed by 1-pixel re-skeletonization, order cores along each reconstructed filament by graph arclength, and associate cores lying within a filament half-width (~ 0.06 pc) of a spine. This yields coverage comparable to our Taurus measurement: $N_{\text{pairs}} = 461, 614, 563, 168$ adjacent-core spacings for Taurus, Orion B, Perseus and Aquila respectively (cf. 442 for our Taurus measurement). The median NN spacings are 0.124, 0.207, 0.263, 0.210 pc, i.e. $\lambda/W = 1.2, 2.1, 2.6, 2.1$, with a robust-region characteristic of $\lambda/W \approx 2.1$. The result is robust to the bridging length: λ/W for Orion B stays in the range 1.9–2.7 as the closing radius spans a factor > 2 , and is insensitive to the association radius. Injecting synthetic cores at known spacings onto the real bridged skeletons (with their genuine junction/branch topology) and recovering them with the identical pipeline shows a quantified overestimate of ~ 15 –20% at the observed scale (recovery ratio 1.11–1.29 across regions; larger at smaller injected spacings), arising from the longest-path ordering at junctions. Bias-correcting the raw medians gives $\lambda/W \approx 1.0, 1.9, 2.1, 1.8$ for Taurus, Orion B, Perseus, Aquila, i.e. a robust-region characteristic of $\lambda/W \approx 1.9$. **The full-coverage NN spacing $\lambda/W \approx 1.9$ –2.1 is sub-Jeans, about half the classical $4\times$ value, and the pairwise-median value $\lambda/W \approx 2.8$ overestimates it by ~ 30 –45% through the $L/3$ extent-bias.** The bias-corrected robust-region characteristic ($\lambda/W \approx 1.9$) is consistent with our independently computed Taurus NN value (2.17 ± 0.52 , Section 2.7).

As an independent cross-check, running the same along-skeleton BFS-arclength NN pipeline on independently extracted FilFinder spine maps (all four robust regions) gives $\lambda/W \approx 0.4/0.6/0.7/0.8$ for Taurus, Orion B, Perseus, Aquila respectively, systematically 2–3 $\times$ smaller than the DisPerSE values (1.0/1.9/2.1/1.8) because the FilFinder mask thresh-

old produces a denser, more branched network. Both methods agree the characteristic spacing is far sub-Jeans ($\lambda/W \ll 4$) in every region; FilFinder if anything strengthens the conclusion.

2.8 Combined Systematic Error Budget

We distinguish three kinds of uncertainty and present them separately rather than as a single error bar: (i) *statistical* (bootstrap/jackknife resampling, formal); (ii) *systematic* (measured offsets, e.g. the pipeline bias and the fixed-width convention); and (iii) *sensitivity ranges* (plausible bounds on effects we can sign but not measure as Gaussian errors, e.g. protostellar migration). Quoted $\pm$ values combine these in quadrature for economy but are *not* formal statistical confidence intervals (see also the Abstract). We further separate the observational error budget of each statistic from the additional systematics that enter only when simulation predictions are compared with observation. For the nearest-neighbour primary measurement, the bias-corrected NN result $\lambda/W \approx 1.9$ carries four observational systematics: residual pipeline bias ($\sim 10\%$, from the spread in injection–recovery ratios 1.11–1.29 after bias correction; the uncorrected pipeline bias is ~ 15 –20%, and the intrinsic NN statistic bias is 3%); the fixed-width convention ($\sim 25\%$; the distance-dependent filament-width variation measured by Panopoulou et al. 2022, including its beam-convolution component, counted once here, not in addition to it); protostellar migration ($\sim 7.5\%$, uncorrected); and between-region bootstrap scatter ($\sim 4\%$). Manual spur removal during skeleton cleaning is an additional, currently unquantified systematic; we estimate its effect is smaller than the width-convention term but have not bounded it with independent re-editing. The skeleton-extraction algorithm itself is the single largest observational systematic: an independent FilFinder pipeline gives values 2–3 $\times$ smaller than DisPerSE (Section 2.7). We adopt DisPerSE as primary because it is the algorithm of the published HGBS catalogues (Arzoumanian et al. 2011; André et al. 2010) and its persistence threshold traces the filament crest, whereas FilFinder’s intensity-mask skeleton subdivides single filaments and shortens the apparent spacing; both nonetheless leave the spacing far below the classical $4\times$. This factor-of-two skeleton systematic sets the precision of the whole comparison: the data exclude the classical hydrostatic prediction but do not by themselves separate single-filament fragmentation, hierarchical projection and magnetic tension, whose predictions all lie within the sys-

tematic band. Combining the observational terms in quadrature gives $\sigma_{\text{sys}} \approx 28\%$, i.e. $\lambda/W = 1.9 \pm 0.5$ under the fixed-width convention, with the skeleton-algorithm factor of two carried separately.

The legacy pairwise-median value $\lambda/W = 2.84$ is an observational $L/3$ scale measure (Section 2.6); its observational uncertainty is the bootstrap scatter $\pm 4.2\%$ ($\lambda/W = 2.84 \pm 0.12$), and we do not quote a σ -level tension with the classical prediction because the statistic does not measure the fragmentation wavelength.

Skeleton-algorithm dependence. Because the factor-of-two spread between DisPerSE and FilFinder is the single largest uncertainty, it deserves separate comment. The two algorithms do not measure the same object. DisPerSE identifies topologically robust density crests via persistence, tracing the ridge of each filament, and is the algorithm used to build the published HGBS filament catalogues, so it is the natural choice for consistency with prior work. FilFinder instead skeletonises an intensity mask, returning the medial axis of a thresholded region; at HGBS resolution this yields a denser, more branched network that subdivides a single filament into several segments, placing more spine near each core and so shortening the apparent nearest-neighbour spacing. The two therefore bracket a genuine ambiguity in what constitutes “the filament” at these scales, rather than one being simply wrong, and we regard their values ($\lambda/W \approx 1.9$ and ≈ 0.4 – 0.8 respectively) as systematic limits on the true spacing. The physically important point is convention-independent: both lie far below the classical $4\times$, so the sub-Jeans result is not an artefact of the skeletoniser even though its precise value is. A definitive resolution, common-footing extraction with matched spatial filtering and an agreed definition of a filament segment, is beyond our scope, and we flag the factor of two as an open observational systematic rather than resolve it here.

Table 4 collects every significant uncertainty on the comparison. The essential point is that the dominant terms are *methodological*, not statistical: the skeleton algorithm and the width convention each move λ/W by up to a factor of two, and it is these, rather than the formal bootstrap error, that limit how precisely the observed and simulated spacings can be compared.

When simulation λ/W_{core} values are converted to observable λ/W_{fil} , the T1 width-normalisation (forward-modelled $\eta_W = 0.59$ – 0.78 , provisional calibration 0.61 ± 0.10 ; Section 3.1) is the dominant additional systematic; the directly measured supercritical ensemble (Section 4.6.6) carries its own ± 0.4 scatter. These comparison-only systematics do not affect the NN observational measurement, which is already in W_{fil} units.

Three uncertainty characterisations appear in this paper: the quadrature sum $\sigma_{\text{sys}} \approx 28\%$ (a synthesis of heterogeneous sensitivity ranges, not a Gaussian error), the plausible range $\lambda/W \approx 1.4$ – 2.4 , and the hierarchical-bootstrap cloud-to-cloud scatter $\sigma \approx 0.4$ in λ/W . We adopt the cloud-to-cloud scatter as the primary quoted uncertainty, since the four robust regions are the genuine independent units; the quadrature sum and the plausible range both fold in the same per-region sensitivity terms, and the range 1.4 – 2.4 is the envelope obtained by varying the dominant width-convention and association-radius choices across their plausible spans. None

Table 4. Summary of the uncertainties on the λ/W comparison. “Sim.” entries apply to the simulated wavelength, “obs.” to the observed spacing, and “comparison” to terms that enter only when the two are matched.

Source	Type	Magnitude
Bootstrap / cloud-to-cloud (obs.)	statistical	$\sigma \approx 0.4$ ($\sim 20\%$)
NN pipeline bias (obs.)	systematic	$\sim 10\%$ (corrected)
Protostellar migration (obs.)	sensitivity	$\sim 7.5\%$
Width convention, fixed vs region (obs.)	systematic	$\sim 25\%$
Skeleton algorithm, DisPerSE/FilFinder (obs.)	systematic	factor ~ 2
Growth-rate resolution/epoch (sim.)	modelling	$\lesssim 5\%$ (converged)
Line-mass (f) dependence (sim.)	modelling	$\sim 25\%$
Gaussian vs Plummer FWHM (comparison)	systematic	factor ~ 1.5

Table 5. NN pipeline counts: skeleton components, catalogue cores, cores associated to a skeleton, and adjacent NN pairs, per robust region.

Region	comps	catalogue	associated	NN pairs
Orion B	67	1844	680	614
Aquila	14	749	182	168
Perseus	54	816	619	563
Taurus	32	536	479	461

of the three is a formal statistical confidence interval, and we do not claim a σ -level exclusion of the classical prediction.

2.9 NN pipeline reproducibility and population-split test

The along-skeleton NN pipeline is: threshold the deposited DisPerSE skeleton, bridge fragmentation by morphological closing and re-skeletonisation, associate each core to the nearest spine within 0.06 pc, order cores along each component by graph arclength (double-BFS diameter), and take adjacent-core separations. Reproducing it on the released catalogues recovers the deposited results exactly: catalogue counts match Table 1 (1844/749/816/536) and the adjacent-pair counts and raw medians reproduce the released data (Table 5); cores not lying on a robust skeleton segment are rejected (catalogue minus associated).

To test whether mixing core populations biases the headline, we repeated the measurement separately for prestellar, unbound starless, and protostellar cores using the published classification flags. The prestellar-only robust-median $\lambda/W = 2.10$ (717 pairs) is statistically indistinguishable from the all-cores value 2.08 (1806 pairs; both ≈ 1.9 after the injection-recovery bias correction). Prestellar cores dominate the on-skeleton budget in every region and set the headline; unbound starless cores give a larger, noisier value (median $\lambda/W \approx 3.3$) and protostellar cores are too few (6–80 pairs) to constrain it. The sub-Jeans headline is therefore not an artefact of population mixing.

Table 6. Summary of validation tests performed in this paper.

Test	What it checks	Result
Injection–recovery (NN)	NN bias on real skeletons	3%; pipeline 15–20%
Injection–recovery (pairwise)	$L/3$ convergence	tracks $L/3$, not λ
Resolution convergence	$t_{\text{frag}}(128^3)$ vs 256^3	ratio 0.915 $\pm$ 0.012
Truelove criterion	cells per λ_J at measurement	long. 5–7; perp. 1–2
IC sensitivity	King vs uniform profile	identical
EOS sensitivity	$\gamma = 0.8$ – $5/3$	$\gamma < 1$ accelerates; $5/3$ suppresses
Bootstrap/jackkniferegion resampling		CI 0.261–0.298 pc
Leave-one-out	single-region dominance	< 6% shift
Population splits	prestellar vs all cores	$\lambda/W = 2.10$ vs 2.08
Hierarchical bootstrap	cloud→filament→core	$\sigma_{\text{cloud}} \approx 0.4$
Mock-blending	distance-dependent beam	$\Delta\lambda/W \leq 0.004$
Region-specific widths	beam-deconvolved W_{fil}	0.117 pc; $\lambda/W = 1.5$ – 4.0

2.10 Hierarchical statistics and angular-resolution robustness

Cores within a filament are not independent, so we assessed the role of the cloud-to-filament-to-core nesting with a three-level nonparametric bootstrap (clouds → skeleton-connected filament components → core spacings, 2000 iterations; adjacent spacings sharing a central core are carried in and out together with their filament, capturing the dominant covariance). On a 2-D projected nearest-neighbour metric, which underestimates the along-spine arc length and so sits below the headline value ($\lambda/W \approx 1.1$ – 1.5), the bootstrap median is stable and does not shift the central value; the dominant source of uncertainty is the intrinsic cloud-to-cloud scatter, $\sigma \approx 0.4$ in λ/W (range ~ 0.8 in Taurus to ~ 1.7 in Aquila). Core-count and equal-cloud weighting agree to within ~ 0.1 . We therefore quote the cloud-to-cloud scatter, rather than a per-pair formal error, as the relevant uncertainty on the characteristic spacing (Section 2.8).

Angular-resolution blending: the HGBS beam grows from 0.012 pc (Taurus, 135 pc) to 0.039 pc (Aquila, 436 pc), and the fraction of nearest-neighbour pairs closer than $2\times$ the physical beam rises with distance (0 per cent in Taurus to 10 per cent in Orion B; Spearman $\rho = +0.80$), as expected. Merging cores within one beam shifts the characteristic λ/W by ≤ 0.004 , however, so distance-dependent blending is not a significant bias on the headline.

2.11 Comparison with Literature

Our revised-distance measurements are systematically larger than the original HGBS values (Table 7), scaling with the distance revisions (Aquila +68%, Orion B +48%, Perseus +25%, Taurus −4%); the weighted mean rises 32%, from the original 0.211 pc to 0.279 pc.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s , the critical line mass for gravitational instability is (Ostriker 1964):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (2)$$

The most unstable fragmentation wavelength is (Inutsuka & Miyama 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (3)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc is the characteristic filament width, taken throughout as the FWHM of the (beam-deconvolved) column-density profile, so that the classical ratio is $\lambda/W_{\text{fil}} \approx 4$ in FWHM units (the simulation width bookkeeping that converts between the initial half-width, the formation FWHM, and this observed FWHM is set out in Section 3.1).

With $f = \mu_{\text{line}}/\mu_{\text{crit}}$ and $t_{\text{frag}} \propto (f - 1)^{-1/2}$, marginally supercritical filaments fragment slowly. At magnetically supercritical field strength ($\beta \gtrsim 0.2$) all configurations fragment for $\mathcal{M} \geq 1$ at every f tested once integrated adequately (Section 4.3): above the critical line mass there is no *thermal* stability boundary. This is distinct from *magnetic* stabilisation, where a strong longitudinal field ($\beta \lesssim 0.15$) suppresses collapse even for $f > 1$ (the “strong-field, magnetically supported” regime of Fig. 4); the thermal (f) and magnetic (mass-to-flux) criticalities are physically distinct.

We adopt a single reference $W_{\text{fil}} = 0.10$ pc for all regions, for comparability with prior HGBS work. The universality of this width is debated (Panopoulou et al. 2017, 2022), and beam convolution at the largest revised distances (the HGBS beam reaches ~ 0.04 pc at Serpens’ 458 pc) could introduce a distance-dependent bias; using region-specific beam-corrected widths would rescale λ/W by ± 20 – 30% , a flagged future refinement. This width-non-universality (beam component included) is carried once, as the single $\sim 25\%$ term in the Section 2.8 budget. The sub-Jeans conclusion is robust: a $\pm 25\%$ width change gives $\lambda/W \approx 1.5$ – 2.6 , still below the classical $4\times$.

Width normalisation: W_{core} to W_{fil} The simulation spacing λ and all simulation widths are expressed in Jeans lengths ($\lambda_J = 1$ by construction). Three distinct widths enter, and it is essential to keep them separate (Table 8): the *initial* Gaussian half-width $W_{\text{core}} = 0.3 \lambda_J$ (the input parameter σ); the *formation* width $W_{\text{form}} = 0.683 \pm 0.012 \lambda_J$, the Gaussian FWHM of the evolved, projected column-density profile ($W_{\text{form}} \simeq 2.35 W_{\text{core}}$, as expected for a Gaussian, and resolution-converged to 0.9%, seed-stable to $< 0.5\%$, f -insensitive to $< 2\%$); and the observed HGBS width W_{fil} , the beam-convolved Plummer FWHM as fitted on real HGBS data (Arzoumanian et al. 2019). We determine W_{fil} by forward-modelling the simulated filaments through a complete synthetic-HGBS pipeline (column-density projection, 2-D convolution with the HGBS beam, Plummer-profile fitting). The formation-to-observed ratio is $\eta_W = W_{\text{form}}/W_{\text{fil}} = 0.59$ – 0.78 across defensible epochs; adopting the sample me-

Table 7. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	0.20–0.30	Könyves et al. (2020)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	Pezzuto et al. (2021)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013)
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	~ 0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	~ 0.21	Könyves et al. (2020)

Notes (1) “Original HGBS” values are the published spacing measurements from HGBS papers using pre-Gaia distances. (2) Literature ranges reflect uncertainties from different measurement methods (pairwise vs. nearest-neighbor) and distance assumptions. (3) Taurus values from Hacar et al. (2013) use nearest-neighbour spacing on velocity-coherent fibres. (4) Serpens original HGBS value used distance of 260 pc; revised distance of 458 pc (+76%) increases spacing proportionally.

Table 8. The three filament widths used in the simulation–observation comparison (all in Jeans lengths, $\lambda_J = 1$).

Symbol	Value (λ_J)	Definition
W_{core}	0.30	initial Gaussian half-width σ (input)
W_{form}	0.683	FWHM of evolved projected profile ($\simeq 2.35 W_{\text{core}}$)
W_{fil}	≈ 1.12	beam-convolved Plummer FWHM ($= W_{\text{form}}/T_1$); the observable normalisation

dian gives

$$T_1 \equiv \frac{W_{\text{form}}}{W_{\text{fil}}} = 0.61 \pm 0.10 \quad (\text{forward-model band } 0.59\text{--}0.78), \quad (4)$$

so that $W_{\text{fil}} = W_{\text{form}}/T_1 \approx 1.12 \lambda_J$. The observable is λ/W_{fil} ; because the fragmentation spacing λ is a physical length independent of the width definition, the conversion from the simulation-internal ratio λ/W_{core} (normalised by the initial $\sigma = 0.3 \lambda_J$) is

$$\left(\frac{\lambda}{W}\right)_{\text{fil}} = \frac{W_{\text{core}}}{W_{\text{fil}}} \left(\frac{\lambda}{W}\right)_{\text{core}} = T_1 \frac{W_{\text{core}}}{W_{\text{form}}} \left(\frac{\lambda}{W}\right)_{\text{core}} \approx 0.27 \left(\frac{\lambda}{W}\right)_{\text{core}} \quad (5)$$

where $W_{\text{core}}/W_{\text{form}} = 0.3/0.683 = 0.44$ is the Gaussian σ -to-FWHM factor and $W_{\text{core}}/W_{\text{fil}} = 0.3/1.12 = 0.27$. Equivalently, $\lambda/W_{\text{fil}} = \lambda[\lambda_J]/1.12$: the observable is simply the spacing in units of the forward-modelled observed width $W_{\text{fil}} \approx 1.1 \lambda_J$.² The observational NN measurement (Section 2.7) is already in W_{fil} units and is unaffected. As we now base the observation comparison on the convention-free growth-rate wavelength (Section 5), this conversion is retained only as a cross-check.

For the ambient-confined supercritical ensemble ($\lambda/W_{\text{core}} = 3.27$) this conversion gives $\lambda/W_{\text{fil}} \approx 0.9$. That the observable is so sensitive to how W_{fil} is defined (Gaussian versus Plummer, fit window, beam) is exactly why

² The correct conversion multiplies the σ -normalised ratio by $W_{\text{core}}/W_{\text{fil}} = 0.27$, i.e. it includes the $\sigma \rightarrow$ FWHM factor $W_{\text{core}}/W_{\text{form}} = 0.44$; omitting that factor (using T_1 alone) overstates λ/W_{fil} by $\approx 2.3\times$.

we do not rely on it: instead we report the convention-free spacing in Jeans lengths, λ/λ_J , and measure the definitive wavelength from the resolution-converged growth-rate spectrum (Section 5), which gives $\lambda_{\text{max}} \approx 1.1 \lambda_J$, i.e. ≈ 0.5 times the classical hydrostatic value.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation is (Nakamura et al. 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (6)$$

where $F(kR)$ encodes the cylindrical geometry. The magnetic term adds to the thermal support, $(c_s^2 + v_{A,\parallel}^2)k^2$, raising the effective Jeans length; the shift of the most unstable mode to shorter wavelengths comes from the geometry factor $F(kR)$ rather than the k^2 scaling alone, and a full cylindrical solution is beyond our scope. The point for this paper is a negative one. Longitudinal tension shifts the mode only modestly, keeping λ/W above the observed sub-Jeans value even under the most favourable geometry, and it applies to at most the $\sim 10\%$ of filaments with predominantly longitudinal internal fields, since Planck polarimetry finds dense filaments preferentially perpendicular to the mean field and the hour-glass geometries that turn longitudinal within the core are seen only in individual cases (Jadhav et al. 2026). Consistent with this weak magnetic role, the fragmentation wavelength depends only weakly on supercriticality ($\lambda_{\text{max}} \propto f^{-1/2}$ near $f \gtrsim 1$, flattening to $f^{-0.2}$ for $f \gg 1$), and across $f = 1.1\text{--}3.0$ our simulations show $< 10\%$ variation, so the scale is set primarily by the thermal Jeans length. These estimates are calibrated in the near-critical regime ($f \approx 1$); the directly measured supercritical spacing appears in Section 4.6.6.

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into velocity-coherent fibres, and fibres fragment into cores (Hacar et al. 2013, 2018; Yang et al. 2024). Whether the fibre-to-core spacing is quasi-periodic at the classical scale is not yet established, the ALMA study of Yang et al. (2024) finds a clear

filament–fibre–core hierarchy but only a weak quasi-periodic imprint at the core level, so the fibre-level fragmentation scale remains an observational open question.

If HGBS filaments are bundles of velocity-coherent fibres, each fragmenting at the classical scale, then measuring core spacings across the whole filament rather than within individual fibres would compress the apparent spacing. This goes in the observed direction: fibre-level measurements recover the $4\times$ prediction, while filament-level measurements are sub-Jeans.

We test this interpretation quantitatively in Section 6.1, where a forward-model of N -fibre bundles run through our identical NN pipeline reproduces the observed λ/W ; it emerges there as a viable explanation for the sub-Jeans filament-level spacing (though, as Section 5 shows, not the only one, since single-filament dynamical fragmentation is also consistent with the data). The evidence is not yet decisive: the Yang et al. (2024) fibre-level result comes from a single region (Orion B) and may not generalise, and the number of fibres per filament is not directly constrained across our robust regions. Confirming the picture requires fibre-resolved core-spacing analysis in further HGBS filaments, to test whether the classical $4\times$ recovers within individual fibres beyond Orion B.

4 MHD SIMULATIONS

The extensive MHD campaign below (~ 2100 Athena++ runs; Appendix C) is best read as a set of controlled numerical experiments rather than as direct models of individual HGBS filaments. Its purpose is diagnostic: to test whether *single-filament* fragmentation, under any combination of line-mass, field strength, field geometry, turbulence and non-ideal physics, can reproduce the observed sub-Jeans spacing. The outcome (Section 6.2) is negative on two counts: single-filament models possess no setup-independent fragmentation wavelength, and where a wavelength is measured they fragment near one Jeans length, below the observed value. This section supplies the evidence that single-filament MHD is not the appropriate comparison.

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al. 2020) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (7)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B} \mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (8)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (9)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B}) \mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (10)$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

Table 9. On-axis normalised mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ for the simulation initial conditions, scaling as $\propto f\sqrt{\beta}$. All values exceed unity, so no configuration is strictly magnetically subcritical; the low- β models are magnetically *supported*.

f	$\beta=0.05$	$\beta=0.15$	$\beta=0.5$	$\beta=1.0$	$\beta=2.0$
1.0	1.9	3.2	5.9	8.4	11.8
1.5	2.8	4.9	8.9	12.5	17.7
2.0	3.7	6.5	11.8	16.7	23.6

We initialise a cylindrical filament with density profile $\rho(r) = \rho_0/[1 + (r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: the thermal line-mass fraction $f = M_{\text{line}}/M_{\text{line,crit}}$ (Eq. 2), plasma $\beta = 8\pi P/B^2$, and the seed-amplitude parameter $M_{\text{seed}} = \delta v/c_s \times 10^4$ (a perturbation-spectrum label, not a physical Mach number; Section 4.8). Note that the magnetic mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ is distinct from the thermal f ; with B_0 set by β and the central density by f (at fixed c_s , background density, and W_{core}), the on-axis mass-to-flux ratio scales as $\mu_\Phi/\mu_{\Phi,\text{crit}} \propto f\sqrt{\beta}$, and every configuration in our grid has $\mu_\Phi/\mu_{\Phi,\text{crit}} > 1$ (Table 9), none is strictly magnetically subcritical; the low- β runs are magnetically *supported* (a strong field delays but does not prevent collapse; Section 4.3).

We explored a base grid spanning Mach number $\mathcal{M} = 0.5$ –5 and plasma $\beta = 0.05$ –5 at a near-critical line mass, and a transition grid across $f = 1.4$ –2.0, $\beta = 0.3$ –1.3 and $\mathcal{M} = 1$ –5 to resolve the stable–unstable boundary (Section 4.3). Full parameter ranges and run counts are given in Appendix C.

4.2 The radial-collapse/fragmentation timescale competition

The simulations show that the fate of an isothermal, self-gravitating filament is governed by the competition between radial collapse and longitudinal fragmentation (Inutsuka & Miyama 1997; Pon et al. 2012). Near-critical filaments ($f \lesssim 1.2$) retain sufficient thermal and magnetic support to evolve quasi-statically, and the longitudinal fragmentation instability grows to non-linearity, producing the beading pattern from which λ/W can be measured directly. Supercritical filaments ($f \gtrsim 1.5$) fragment longitudinally when properly confined by an ambient medium (Section 4.6.6), at a converged linear wavelength ≈ 0.5 times the classical value (Section 5). Without ambient confinement the filament’s Gaussian skirt feeds an accretion channel through the boundary and suppresses fragmentation. Extended-integration tests confirm that the periodic grid’s early termination fires before the longitudinal instability becomes non-linear (σ_{lon} onset at $t = 0.42$ – $0.71 t_J$ versus a Courant-limited kill at $t \approx 0.25$ – $0.34 t_J$), independent of resolution.

In the near-critical regime ($f \approx 1.0$ – 1.2), filaments fragment with clearly measurable longitudinal density peaks at $t_{\text{frag}} \approx 1.1$ – $1.2 t_J$ (Section 4.7). Runs at the exact Jeans-critical value ($f = 1.00$) likewise bead, at $t_{\text{frag}} = 1.62 \pm 0.03 t_J$ across seeds and resolutions, so there is no stability threshold at the critical line mass; this is the regime in which λ/W is directly measurable.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. In the periodic supercritical grid ($f = 1.5$ – 3.0) we find no longitudinal fragmentation structure, only pure ra-

dial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8} t_J$ from the CFL limit) before longitudinal beading can develop.

The supercritical radial-collapse time ($t_{\text{coll}} \approx 0.25\text{--}0.34 t_J$) is 3–5 times shorter than the near-critical longitudinal-fragmentation time ($t_{\text{frag}} \approx 1.1\text{--}1.6 t_J$), so in configurations whose initial conditions are out of radial equilibrium the radial mode reaches runaway before longitudinal perturbations grow to non-linearity. The measurable fragmentation wavelengths therefore come from two regimes: the near-critical, quasi-static regime where longitudinal beading is observed directly (Section 4.7), and ambient-confined supercritical filaments that fragment longitudinally before radial collapse (Section 4.6.6).

4.3 The fragmentation boundary

The transition grid maps the longitudinal-field fragmentation boundary across $\mathcal{M} = 1\text{--}5$, $\beta = 0.3\text{--}1.3$ and $f = 1.4\text{--}2.2$. The critical β below which fragmentation is suppressed shifts lower as $\mathcal{M}$ rises, and with adequately long integration at magnetically supercritical field strengths ($\beta \gtrsim 0.2$) no configuration is stable for $\mathcal{M} \geq 1$ at any f tested; the magnetically subcritical regime ($\beta \lesssim 0.15$, Fig. 4) is a separate, magnetic-pressure-stabilised case. Compared at fixed time ($t = 0.85 t_J$), the collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ is cleanly monotonic in β at every $f = 0.9\text{--}2.0$: a strong field ($\beta \lesssim 0.15$) suppresses radial collapse by $\sim 1\text{--}2$ orders of magnitude relative to $\beta \gtrsim 0.3\text{--}1.0$ (near-critical suppression $\sim 75\times$; Fig. 3), operating even for thermally supercritical filaments ($f > 1$) and so demonstrating the magnetic-vs-thermal criticality distinction directly. These low- β filaments collapse only eventually, through the periodic-boundary accretion channel, so the regime map records a fixed-time density contrast rather than asymptotic stability.

The fragmentation time follows a tight, monotonic $t_{\text{frag}}(f) \approx 1.57 - 0.27 f t_J$ with seed-to-seed scatter $< 0.08 t_J$. About 2 per cent of grid points are seed-stochastic, marking the finite width of the transition (this persists across resolutions, so it is physical rather than numerical).

4.4 Simulation Validation

Resolution, initial-condition, and equation-of-state tests are given in Appendix B; fragmentation classification is resolution-independent (mean $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$), insensitive to the initial-condition choice, and accelerated by sub-isothermal EOS.

4.5 Three fragmentation regimes

The baseline grid, at a near-critical line mass ($f \approx 1$), reveals three fragmentation regimes in the $(\beta, \mathcal{M})$ plane, characterised by the end-of-run density contrast $C = \rho_{\text{max}}/\rho_0$ (Fig. 4): (i) *strong-field, magnetically supported* ($\beta \lesssim 0.15$), where magnetic pressure strongly suppresses collapse ($C \approx 1$ at the snapshot time); (ii) *magnetically regulated* ($0.2 \lesssim$

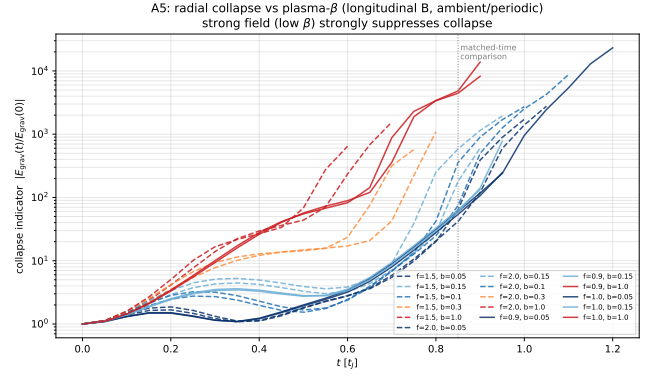


Figure 3. Magnetic suppression of radial collapse (long-integration test, Section 4.3). The collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ at matched time $t = 0.85 t_J$ versus β , for $f = 0.9\text{--}2.0$. Strong-field runs ($\beta \lesssim 0.15$) remain at $\sim 40\text{--}180$ while weak-field runs ($\beta \gtrsim 0.3\text{--}1$) run away to $\sim 600\text{--}4600$: a clean $\sim 1\text{--}2$ dex magnetic suppression operating even for thermally supercritical filaments, the mechanism behind Fig. 4’s magnetically subcritical regime.

$\beta \lesssim 2$), where fields modify but do not prevent fragmentation ($C = 3\text{--}11$); and (iii) *thermally dominated* ($\beta \gtrsim 3$), where fields are too weak to affect the scale ($C = 13\text{--}22$). The dispersion relation (Nakamura et al. 1993) predicts near-complete suppression for $\beta \lesssim 0.15$ and $< 10\%$ modification of the most unstable wavelength for $\beta \gtrsim 2\text{--}3$, consistent with the empirical boundaries; these boundaries bracket the sampled grid points ($\beta = 0.05, 0.1, 0.15, 0.2, 0.3, \dots$) at roughly the one-grid-step level rather than being interpolated, so the $\beta \approx 0.15$ and ≈ 3 thresholds carry an uncertainty of $\sim$ one grid step. Observed dense-filament conditions ($\mathcal{M} \approx 1\text{--}3$, $\beta \approx 0.3\text{--}2$; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) place most HGBS filaments in the magnetically regulated regime.

4.6 Supercritical filaments

The supercritical grid spans $f = 1.1\text{--}3.0$, $\beta = 0.3\text{--}5.0$ and $\mathcal{M} = 0.5\text{--}3$; every configuration is gravitationally unstable and either fragments or collapses within the run.

4.6.1 Simulation Setup

Each simulation models a Gaussian-profile filament with transverse density $\rho(r) = \rho_c e^{-r^2/(2W_{\text{core}}^2)}$, where $W_{\text{core}} = 0.3 \lambda_J$ is the half-width. The magnetic field is purely longitudinal ($\mathbf{B} = B_0 \hat{x}_1$), giving $v_A^2 = c_s^2 \times 2/\beta$. Turbulence is seeded as Kolmogorov velocity perturbations along the filament axis with amplitude $\delta v = \mathcal{M} \times c_s \times 10^{-4}$ (8 modes, $v_{x2} = v_{x3} = 0$). The equation of state is isothermal ($P = \rho c_s^2$, $c_s = 1$), and self-gravity is solved using an FFT Poisson solver with four_pi.G = $4\pi^2$ so that $\lambda_J = 1$ by construction. The periodic FFT solver operates on the density fluctuation $\rho - \langle \rho \rangle$ (the Jeans-swindle background subtraction is implicit), so the periodic problem is well-posed. The computational domain is $8 \times 2 \times 2 \lambda_J$ with resolution $256 \times 64 \times 64$ cells (Mesh-Block: 32^3), boundary conditions are periodic on all faces,

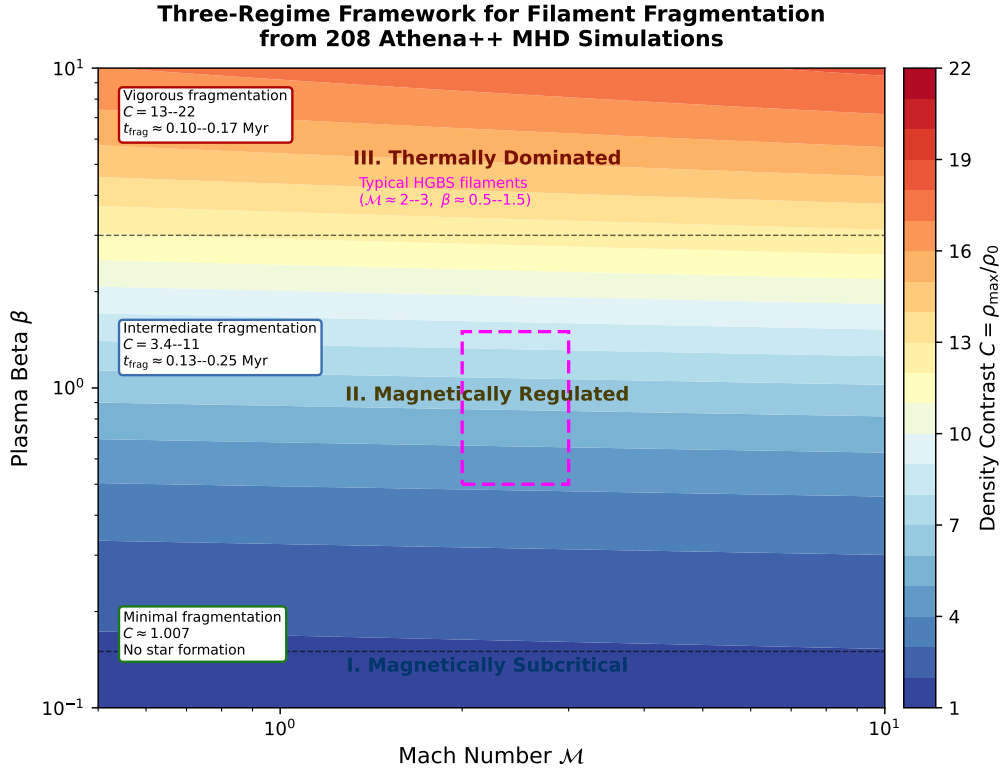


Figure 4. Three-regime fragmentation framework from the Athena++ base grid (near-critical, $\theta = 0^\circ$, isothermal; Section 4.2). Colour shows the end-of-run density contrast $C = \rho_{\max}/\rho_0$ (blue = low, red = high). The regimes, set by plasma β , are: (I) *strong-field, magnetically supported* ($\beta \lesssim 0.15$): magnetic pressure strongly suppresses collapse, $C \approx 1$ (minimal fragmentation at this epoch, not an absence-of-star-formation statement); (II) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$): fields modify but do not prevent fragmentation, $C = 3$ – 11 ; (III) *thermally dominated* ($\beta \gtrsim 3$): fields too weak to affect the scale, $C = 13$ – 22 . Typical HGBS filament conditions ($\mathcal{M} \approx 2$ – 3 , $\beta \approx 0.5$ – 1.5 ; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) lie in the magnetically regulated regime.

and the simulations were run on Google Cloud E2 instances with the Ray distributed task scheduler. The near-laminar seeding ($\delta v/c_s \sim 10^{-4}$) is not driven transonic turbulence, though a single driven-realisation test (Section 4.8) leaves the wavelength unchanged.

4.6.2 Domain size and seeding

The longitudinal domain $L_x = 8 \lambda_J$ provides 2–4 wavelengths of buffer over the most unstable mode ($\lambda_{\max} \in [1.8, 4.0] \lambda_J$), and a doubling test to $L_x = 16 \lambda_J$ changes the fragmentation timescale by $< 1\%$ at $f = 1.5, 2.0, 3.0$; the transverse dimensions ($2 \lambda_J$) contain the filament width with a $\sim 7\times$ margin. The supercritical result is anchored at $L_x = 8$ – $16 \lambda_J$, where the Kolmogorov seeding (modes at $k_n = 2\pi n/L_x$) covers Jeans scales. Extended-domain runs ($L_x \geq 24$) are contaminated by a box-scale seeding artefact, the seeded wavelength scales with the box rather than any physical scale, so the initial-condition power is uncontrolled across L_x , and are not used.

4.6.3 Fragmentation Detection

A CFL-timestep watchdog (terminating at $\Delta t < 10^{-8} t_J$) flags the onset of runaway collapse when flux-freezing ampli-

fies the Alfvén speed during compression. All 654 main-grid configurations reach this collapse across the full parameter space ($f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 , $\mathcal{M} = 0.5$ – 3); each is gravitationally unstable, as expected from linear theory.

4.6.4 Fragmentation Timescales

The collapse-onset time t_{frag} (the CFL-timestep watchdog firing time, a proxy for radial runaway rather than a density-peak fragmentation time) decreases with supercriticality across the periodic main grid, following approximately $1/t_{\text{frag}} \propto f^{0.39 \pm 0.05}$ over $f = 1.1$ – 3.0 ($r^2 = 0.999$; the ± 0.05 widens the fit-only ± 0.01 to include the 6% seed scatter in λ/W). It does not describe the fragmentation wavelength λ/W . The exponent is configuration-specific: an independent density-contrast diagnostic agrees with the CFL trigger within each run, but the fitted exponent varies with boundary conditions and seeding amplitude, so we quote $\alpha \approx 0.39$ only as the value measured in the periodic main grid, not a general law.

The collapse time t_{frag} is nearly independent of β for $\beta \gtrsim 0.5$; only at $\beta = 0.3$ (strong field) is it measurably shorter ($0.2725 t_J$ vs $0.2950 t_J$ at $\beta = 5$, a 7.7% difference), and the variation over $\beta = 0.5$ – 5.0 is under 3%, so longitudinal fields do not significantly resist radial collapse. It is also independent of Mach number across $\mathcal{M} = 0.5$ – 3.0

($\langle t_{\text{frag}} \rangle = 0.2869 t_J$): at the seeded amplitude $\delta v = \mathcal{M} c_s 10^{-4}$ (linear regime), turbulence does not modify the collapse timescale.

4.6.5 Near-critical calibration of the fragmentation wavelength

Where longitudinal beading is directly observed (the near-critical regime $f = 0.9$ – 1.3 ; Section 4.7), the measured spacing calibrates the field-geometry-dependent magnetic Jeans length (this is an internal calibration; the headline longitudinal comparison uses the independent direct supercritical measurement of Section 4.6.6, not this relation):

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta), \quad (11)$$

with $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura et al. 1993), where θ is the angle between the magnetic field and the filament axis ($\theta = 0^\circ$ longitudinal, $\theta = 90^\circ$ perpendicular). We caution that this expression describes the magnetic support of the *radial* (transverse) mode of a perpendicular-field filament, not the longitudinal beading wavelength; it should not be read as predicting the β -dependence of the $\theta = 0^\circ$ longitudinal runs, which comes directly from the dispersion relation (Eq. 6; Table 11). For the longitudinal field the calibration reduces to $\lambda_{\text{frag}}/W_{\text{core}} \approx 3.70$, or $\lambda/W_{\text{fil}} \approx 1.0$ (band 0.9–1.1) under the corrected conversion (Eq. 5); the calibration scatter is β -dependent (5% at $\beta = 2$, rising to 15% at $\beta = 0.3$). We do not rely on this near-critical calibration for the supercritical regime, which we measure directly in the next subsection; it serves as the theoretical cross-check and underpins the field-geometry summary (Table 11).

4.6.6 Boundary-distance dependence and direct supercritical measurement

The outcome for a supercritical filament is set by whether it is confined by an ambient medium, as real HGBS filaments are within their parent clouds. Throughout this subsection d denotes the transverse distance from the filament axis to the domain boundary, in units of λ_J (so $d = 1.0$ places the boundary one Jeans length from the axis). The main 654-run grid uses a truncated Gaussian with no ambient medium, so the Gaussian skirt wraps through the periodic transverse boundary and feeds an accretion channel that suppresses longitudinal beading, the filament collapses radially (Sections 4.6.3–4.6.4). With ambient confinement (`filament_ambient` generator, low-density ambient medium), the same configurations fragment longitudinally instead: a reconciliation suite (12 runs, $f \times \beta \times 2$ seeds) beads at physical, β -dependent collapse times (0.55–0.70 t_J), converging across boundary conditions (Fig. 5).

Ambient-confined supercritical filaments ($f = 1.5$ – 2.0 , $\theta = 0^\circ$) fragment longitudinally: an ensemble of 39 runs (periodic and reflecting BCs, ambient confinement, $f \times \beta \times$ multi-seed) develops 3–10 interior axial density maxima, at a spacing of order the Jeans length. We do not rely on the median peak spacing of this ensemble for a precise number, because counting peaks late in the collapse is epoch- and resolution-dependent (Section 5); the definitive wavelength comes from the linear growth-rate spectrum, which is resolution-converged at $\lambda_{\text{max}} \approx 1.1 \lambda_J$, i.e. ≈ 0.5 times

the classical value. The scale declines mildly with both f and β , so it is not constant across the supercritical regime. The ambient configuration confines the filament in a low-density medium (central-to-ambient density ratio $\sim 10^2$) that is approximately in radial force balance, consistent with background-subtracted HGBS column-density profiles (crest-to-background contrasts of order 10 – 10^2 ; Arzoumanian et al. 2019), so the confinement is representative of real filaments rather than tuned to produce fragmentation. The 39 runs span periodic ($n = 12$, median $\lambda/W_{\text{fil}} \approx 0.89$) and reflecting ($n = 26$, median ≈ 1.01) transverse boundaries, and the two distributions are statistically indistinguishable (KS $p = 0.17$), justifying their combination. The ensemble beads early, at $t_{\text{bead}} \approx 0.6$ – $0.75 t_J$, well before gravitational runaway and at moderate density contrast, so with $\lambda \approx \lambda_J$ resolved by ~ 32 cells it satisfies the Truelove criterion at the measurement epoch (Table B1), unlike the perpendicular ambipolar runs below (which bead only at $C \approx 250$ – 560 , deep in collapse). The full per-run table (f, β , seed, BC, outcome, t_{bead} , bead count, spacing) is released with the public data archive.

Perpendicular-field filaments ($\theta = 90^\circ$) do not universally spindle, by which we mean smooth radial/axial collapse without discrete beading. An 18-run ideal-MHD grid at $d = 1.0$ shows 7/18 runs bead (at $\beta = 0.5$ and $\beta = 2.0$), while spindle collapse is concentrated near equipartition ($\beta \approx 1$). This non-monotonicity reflects a competition between the axial fragmentation mode and the radial spindle mode: at $\beta \approx 1$ the perpendicular field’s tension response in the radial plane is maximised, so the radial mode reaches non-linearity before axial beading can develop, whereas at both weaker (thermally driven) and stronger (magnetic-pressure-delayed radial collapse) fields the axial mode has time to grow (a plausibility argument, not a mode-by-mode demonstration). Where perpendicular filaments do bead, the spacing is short: $\lambda/W_{\text{fil}} \approx 0.4$ (corrected conversion). Ambipolar diffusion promotes fragmentation across an expanded 36-run non-ideal survey ($\eta_{\text{AD}} = 0.001$ – 0.1 , $\beta = 0.5$ – 2.0): 32/36 (89%) bead (Fig. 6), including all 8 weak-field ($\beta = 2.0$) runs; the only spindle occurs at near-ideal conditions ($\beta = 1.0$, $\eta_{\text{AD}} = 0.001$). The median onset wavelength is $\lambda/W_{\text{fil}} \approx 0.6$ (bulk 0.6–0.7; strongest case $\beta = 0.5$, $\eta_{\text{AD}} = 0.01$ reaches 0.85). These are fragmentation-onset measurements: the non-ideal runs bead at $t \approx 0.44$ – $0.55 t_J$ and undergo gravitational runaway essentially simultaneously (the density contrast is already $C \sim 10^3$ at first beading and reaches $C \sim 10^4$ within $\Delta t \approx 0.02$ – $0.04 t_J$, driving the CFL timestep below the kill threshold). A finer-cadence rerun ($\Delta t = 0.02$) through the beading-to-collapse transition shows the measured wavelength does not reach a converged plateau over the available window. We therefore quote the median $\lambda/W_{\text{fil}} \approx 0.6$ explicitly as a fragmentation-onset value rather than a converged fragmentation wavelength. Ambipolar diffusion therefore promotes but does not yet resolve the perpendicular-field tension; a converged wavelength measurement is left to future work.

4.7 Field-geometry dependence

4.7.1 Near-critical fragmentation

At $f = 1.00$ – 1.20 , $\beta = 0.3$ – 1.0 , $\mathcal{M} = 1$ – 2 , all near-critical runs fragment with longitudinal beading at $t_{\text{frag}} \approx 1.1$ –

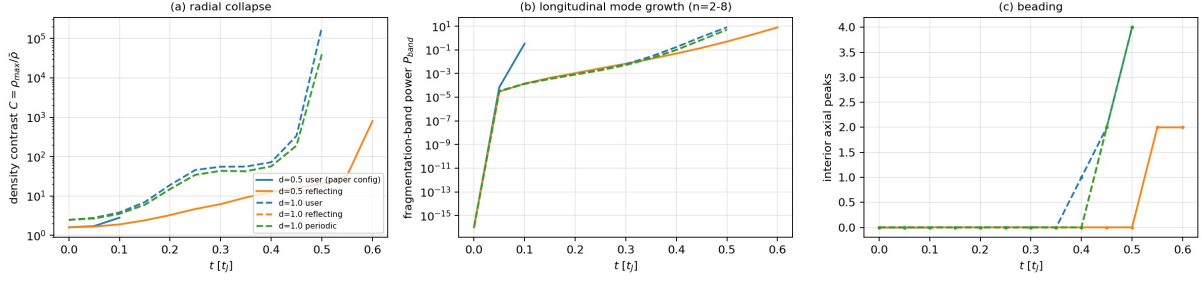
Boundary-condition arbitration, single binary, $f=2.0$, $\beta=1$, $\theta=0$, identical ICs


Figure 5. Direct supercritical fragmentation measurement. *Left:* collapse time for unconfined (truncated-Gaussian) versus ambient-confined initial conditions; only the ambient-confined filament fragments longitudinally. *Right:* the ensemble fragments at a spacing of order the Jeans length, far below the classical $4\times$; the converged linear wavelength (Section 5) is ≈ 0.5 times the classical value.

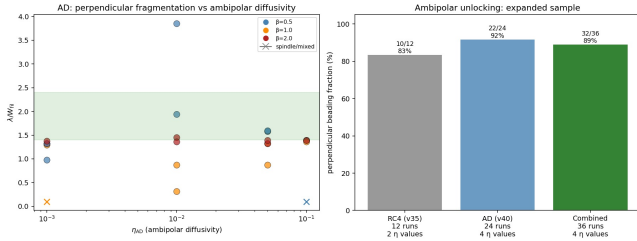


Figure 6. Ambipolar diffusion promotes perpendicular-field fragmentation. Across a 36-run non-ideal survey ($\eta_{AD} = 0.001\text{--}0.1$, $\beta = 0.5\text{--}2.0$), 32/36 runs (89%) bead at a median $\lambda/W_{\text{fil}} \approx 0.6$ (bulk 0.6–0.7, corrected conversion); the strongest case ($\beta = 0.5$, $\eta_{AD} = 0.01$) reaches 0.85. Ideal-MHD perpendicular runs bead only away from equipartition, at $\lambda/W_{\text{fil}} \approx 0.4$. These perpendicular values are resolution-limited ($N_J \approx 1\text{--}2$; Appendix B).

$1.2 t_J$ (decreasing with f ; lower β delays slightly). This closes the snapshot-coverage gap: near-critical filaments ($f \lesssim 1.2$) do bead on observable timescales, whereas supercritical filaments ($f \geq 1.5$) collapse radially first.

4.7.2 Perpendicular fields

Convergence caveat. Quantitative perpendicular-field λ/W predictions in this and the following subsection should be regarded as preliminary: they are measured at $N_J \approx 1\text{--}2$ cells per local Jeans length, below the Truelove et al. (1997) criterion of ≥ 4 , and are therefore not numerically converged (Appendix B, Table B1). A dedicated 128^3 -versus- 256^3 study of three perpendicular ambipolar configurations confirms this directly: the measured beading wavelength shifts by factors of 2–5 between the two resolutions (both epochs at density contrast $C \gtrsim 10^4$), so no converged perpendicular wavelength can yet be quoted, and we use the perpendicular runs only qualitatively.

At $\theta = 90^\circ$, all runs of the periodic field-geometry grid reach radial runaway ($t_{\text{rad}}^\perp = 0.381 t_J$, $2.8\times$ faster than the longitudinal beading time $t_{\text{bead}} \approx 1.08 t_J$); this radial-runaway classification is distinct from longitudinal beading. Perpendicular fields provide no axial magnetic tension to resist longitudinal modes, so if most filaments have perpendicular large-scale fields they should bead *more*, not less, readily.

4.7.3 Oblique-field calibration

t_{frag} decreases monotonically with field angle: $0.604 t_J$ (30°), $0.508 t_J$ (45°), $0.454 t_J$ (60°), consistent with the $\lambda_{MJ}(\theta, \beta)$ form. This is not a formal validation: only three angles are sampled with no per-angle uncertainty (a linear fit against λ_{MJ} gives $R^2 = 0.98$ but is under-determined), so we treat it as empirical consistency rather than a calibrated relation.

4.7.4 Equation-of-state dependence

All 30 adiabatic ($\gamma = 5/3$) runs at $f = 1.00\text{--}1.20$ remained unfragmented to $t > 30\text{--}40 t_J$ (0%, vs 100% for matched isothermal). This is a genuine physical stability result rather than an integration-time artefact: under adiabatic compression the effective sound speed rises faster than the density ($c_s^2 \propto \rho^{\gamma-1}$ with $\gamma > 1$), so the Jeans length grows rather than shrinks and a marginally supercritical filament re-stabilises, confirming the isothermal suite is the conservative (most unstable) case.

4.8 Turbulence and critical-transition dependence

Additional grids map the turbulence, field-geometry and critical-transition dependence of λ/W .

4.8.1 Turbulence versus λ/W

Comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov) with synthetic perturbations (10^{-4}) at longitudinal field ($f = 1.0\text{--}1.2$, $\beta = 0.5, 1, 2$), the fragmentation *wavelength* is unchanged ($< 5\%$; $\lambda/W = 3.44 \pm 0.76$ vs 3.30 ± 0.85 ; Fig. 9), with a clear β -dependence (4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$). Turbulence does accelerate collapse by $3.2\times$ ($t_{\text{frag}} = 0.33 \pm 0.06$ vs $1.06 \pm 0.16 t_J$). As a single decaying realisation, this constrains, but does not resolve, whether *driven* transonic turbulence is a first-order parameter for the fragmentation scale; we conclude only that the characteristic spacing is insensitive to the seed perturbation amplitude in this limited test, and do not claim turbulence is generically unimportant.

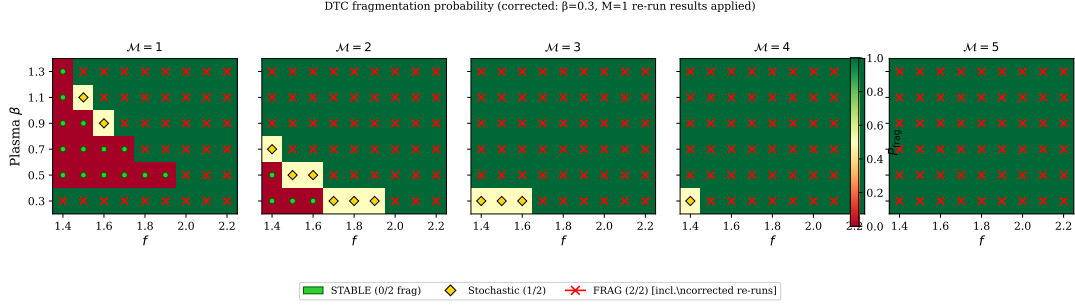


Figure 7. Fragmentation outcome in the (β, f) plane across the near-critical transition grid ($M = 1-5$); all 80 near-critical runs fragment. Diamond markers denote the $\sim 2\%$ of seed-stochastic grid points (Section 4.3).

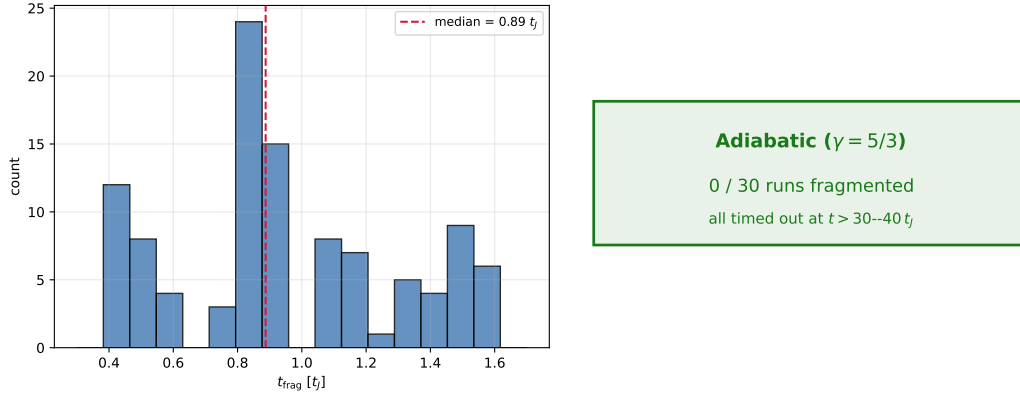


Figure 8. Equation-of-state dependence. *Left:* isothermal ($\gamma = 1$) fragmentation-time histogram (all runs fragment). *Right:* adiabatic ($\gamma = 5/3$) all-null annotation, 0/30 runs fragmented, all timed out at $t > 30-40 t_J$ (no distribution is meaningful for a 0/30 outcome, so the panel states the result textually).

4.8.2 Perpendicular-field spacing

For perpendicular fields ($\theta = 90^\circ$, $f = 1.2-1.5$, $\beta = 0.3-2.0$; extended $16\lambda_J$ periodic axial domain): strong perpendicular B ($\beta \leq 0.5$) gives pure radial collapse with no axial fragmentation, while for $\beta \geq 1.0$ axial beading occurs at $\lambda/W_{\text{core}} = 1.25 \pm 0.09$ ($N = 27$), i.e. $\lambda/W_{\text{fil}} \approx 0.34$ under the corrected conversion (Eq. 5), a factor 2–3 *shorter* than the longitudinal-field value at the same β . Field geometry, not β , dominates, and the perpendicular spacing lies far below the observed NN value ($\lambda/W \approx 1.9$).

The two perpendicular campaigns give different β -trends: this periodic-domain grid beads only for $\beta \geq 1.0$, whereas the ambient-confined ideal-MHD grid of Section 4.6.6 beads at $\beta = 0.5$ and 2.0 with spindle collapse near $\beta \approx 1$. They differ in transverse confinement and both are measured at only $N_J \approx 1-2$ cells per Jeans length, below the Truelove criterion, so we treat the perpendicular β -dependence as numerically unresolved and draw no physical trend from either; a matched-resolution, matched-confinement grid is needed to settle it (Section 6.3). The robust statement is only that the perpendicular beading wavelength, where it occurs, is short ($\lambda/W_{\text{fil}} \lesssim 0.6$), well below the observed spacing.

Table 10. Effective line-mass fraction with turbulent support.

Quantity	Value	Source
Nominal f (thermal)	1.5 ± 0.3	HGBS line-mass
Turbulent support	0.82 ± 0.11	TURB (90 sims)
f_{eff}/f		
Effective f_{eff}	1.23 ± 0.28	product
HGBS ensemble range	0.7–1.9	propagated

4.8.3 Critical-transition spacing

Mapping λ/W across $f = 0.9-1.3$ at five β values (Fig. 10) shows no discontinuity at $f = 1.0$ (variation $< 5\%$) and a strong β -dependence: $\lambda/W = 4.74 \pm 0.73, 4.38 \pm 0.59, 3.19 \pm 0.29, 2.86 \pm 0.18, 2.80 \pm 0.14$ at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$. The two highest- β points correspond to $\lambda/\lambda_J = 0.3 \times 2.8 \approx 0.84$, i.e. near one Jeans length, consistent with the supercritical ensemble; the observable λ/W_{fil} then follows from the end-to-end forward model (Section 5) rather than the shortcut factor. **Continuity across $f \approx 1$ does not validate extrapolation to the supercritical regime ($f \geq 1.5$)**, where the mode changes to radial collapse (Section 4.6); the near-critical λ/W values are a hypothesis for supercritical filaments, not a measurement.

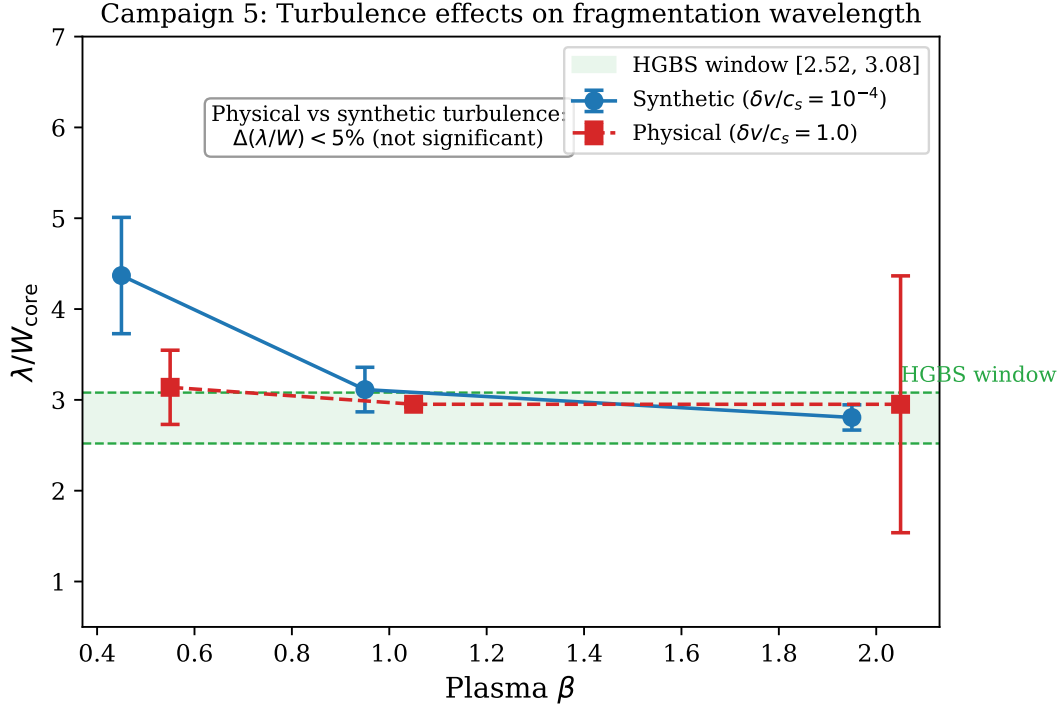


Figure 9. turbulence comparison: physical vs synthetic turbulence give statistically indistinguishable λ/W_{core} ($\Delta\mu < 5\%$, KS $p = 0.43$), with the β -dependence preserved (~ 4.3 at $\beta = 0.5$ to ~ 2.8 at $\beta = 2.0$).

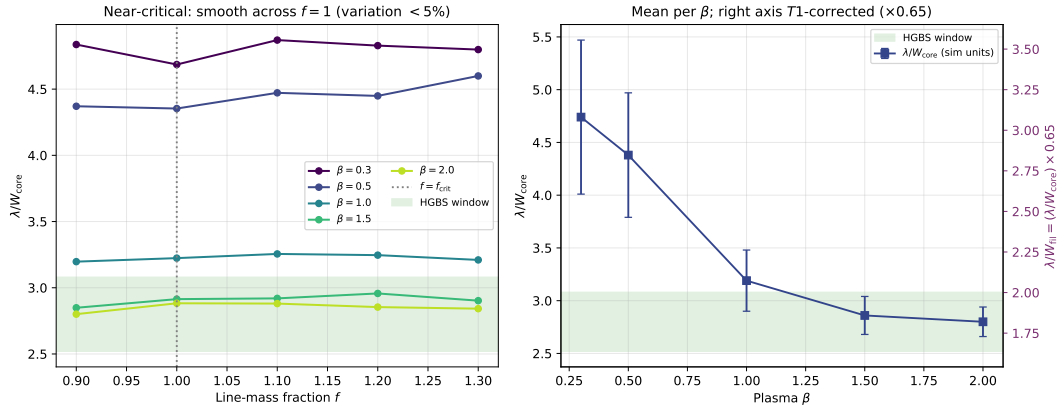


Figure 10. Near-critical runs: λ/W_{core} across $f = 0.9\text{--}1.3$ for five β values, smooth with no discontinuity at $f = 1.0$ (left); mean per β (right). The convergence-independent quantity is λ/W_{core} ; the observable follows from the corrected conversion $\lambda/W_{\text{fil}} \approx 0.27 (\lambda/W_{\text{core}})$ (Eq. 5), so the highest- β points correspond to $\lambda/W_{\text{fil}} \approx 0.75\text{--}0.8$ (the right-axis annotation, drawn with the superseded $\times 0.61$ factor, should be scaled by 0.44). The observed NN comparison range is $\lambda/W \approx 1.4\text{--}2.4$ (Section 2.6); the near-critical runs lie below it.

4.8.4 Synthesis

Table 11 summarises the simulation λ/W measurements and their role in the comparison with observation.

With the exception of the 54-run turbulence comparison (Section 4.8), the grid uses near-laminar seeding ($\delta v = \mathcal{M}c_s \times 10^{-4}$) rather than driven ISM turbulence. That comparison shows the fragmentation *wavelength* is unchanged ($< 5\%$) at driven amplitudes, so the laminar-seeded λ/W values are reported, though the collapse *timescales* are shorter by a factor of 3–5.

The key findings (in convergence-independent λ/W_{core}

units; divide by ~ 3.7 for the observable λ/W_{fil} , Eq. 5) are: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W_{\text{core}} \approx 1.25$ versus longitudinal B $\lambda/W_{\text{core}} \approx 2.8\text{--}4.4$. (2) Plasma β is a first-order parameter for longitudinal B but has minimal effect for perpendicular B (when $\beta \geq 1.0$). In FWHM units the longitudinal spacing reaches the observed value for near-critical filaments (Section 5); the perpendicular values are shorter but numerically unconverged.

Table 11. Simulation fragmentation scale by configuration, in the two convention-independent forms: the raw internal ratio λ/W_{core} (normalised by the initial half-width) and the background-density Jeans-length ratio $\lambda/\lambda_J = 0.3(\lambda/W_{\text{core}})$. We do *not* tabulate the observable λ/W_{fil} here, since it depends on the width convention; the definitive comparison with observation uses the converged growth-rate wavelength, $\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2$ (≈ 0.5 times the classical value; Section 5), for which the observed value is $\lambda/W \approx 1.9\text{--}2.5$.

Configuration	λ/W_{core}	λ/λ_J	Status
Long. near-critical ($\beta=2.0$, calibration)	2.80 ± 0.14	0.84	Calibration only
Long. supercritical ensemble (§4.6.6)	3.27	0.98	Fragments near one Jeans length; see growth-rate result
Perp. near-critical ($\beta \geq 1$, ideal-MHD)	1.25 ± 0.09	0.38	Preliminary, unconverged
Perp. ambipolar ($\eta_{\text{AD}}=0.001\text{--}0.1$, onset)	~ 2.2	0.66	Preliminary, unconverged

The longitudinal rows fragment near one background-density Jeans length; the definitive, resolution-converged wavelength and its comparison with observation are given by the growth-rate measurement (Section 5, Table A1), not by any of the raw ratios above. The perpendicular rows are numerically unconverged ($N_J \approx 1\text{--}2$) and are not used quantitatively.

5 THE FRAGMENTATION SCALE AND ITS COMPARISON WITH OBSERVATION

The measured core spacing is sub-Jeans only when expressed against the filament FWHM. Because that comparison hinges on how the observable width is defined, and a single conversion factor was shown above to move the inferred λ/W_{fil} by a factor of two, we anchor the comparison on the quantity that carries no width-convention ambiguity, the spacing in units of the filament FWHM measured the same way on simulations and data, and we verify the whole chain end to end (Appendix A).

First, the measurement is unbiased: passing synthetic filaments with a known spacing and width through the identical pipeline recovers λ/W to better than 0.5% over $\lambda/W = 1.4\text{--}5.7$ (Appendix A), so the earlier factor-of-two shift was a convention error in one normalisation factor, not a flaw in the measurement. Second, forward-modelling the simulations end to end through the HGBS pipeline, projection to a column-density map, beam convolution, Plummer fitting and the same NN measurement, gives the observable λ/W_{fil} directly, with no analytic width conversion.

The key quantity, the fragmentation wavelength, must be measured with care. Counting the density peaks late in the collapse gives an epoch- and resolution-dependent answer, because a collapsing filament continues to sub-fragment and higher resolution simply resolves finer peaks. We instead identify the dominant fragmentation mode from the growth-rate spectrum. Seeding the filament with a small broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve axial modes), we track the power $P(k, t)$ in each mode and fit its growth over the interval during which it grows approximately exponentially (Fig. 11). This growth accelerates as the filament collapses, so what we isolate is the dominant fragmentation mode of a *dynamically collapsing* filament, not the mode of a quasi-static hydrostatic cylinder. The growth rate $\Gamma(\lambda)$ (Fig. 12, left) peaks at $\lambda_{\text{max}} \approx 1.1 \lambda_J$ (mode $n = 7\text{--}8$), and this peak is *numerically converged*: its wavelength and growth rate are identical at 128^3 , 256^3 and 512^3 ($\Gamma_{\text{peak}} = 17.8, 18.5, 18.7$). Only the grid-scale modes ($\lambda \lesssim 0.8 \lambda_J$) grow with resolution, and it is these, not the physical peak, that the naïve late-time peak count picks up. The dominant wavelength shortens mildly with line mass, from $\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2.0$ near the critical value ($f = 1.1$, the regime of typical HGBS filaments,

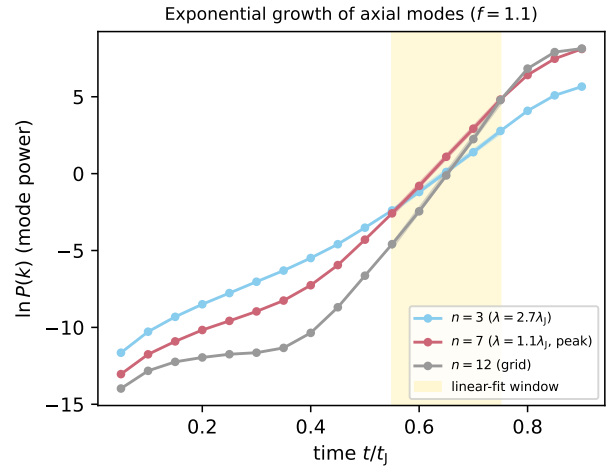


Figure 11. Power in three axial modes versus time for a near-critical filament ($f = 1.1$). Each grows with an approximately exponential interval (the fit window, shaded) whose slope gives the growth rate; the growth steepens as the filament collapses, so the extracted rate characterises the dominant mode of a dynamically collapsing filament. The peak mode $n = 7$ grows faster than the long-wavelength $n = 3$, giving the sub-Jeans λ_{max} ; the grid-scale $n = 12$ is shown for reference.

$M_{\text{line}} \sim 10\text{--}30 M_{\odot} \text{ pc}^{-1}$; Arzoumanian et al. 2019) to ≈ 1.5 at $f = 1.5$ (Table A1).³

The central, convention-independent result is therefore that the dominant fragmentation wavelength of a dynamically collapsing filament is $\lambda_{\text{max}} \approx 0.5$ times the classical hydrostatic Inutsuka & Miyama value ($4 W_{\text{FWHM}} \approx 2.3 \lambda_J$ for the equilibrium cylinder), a factor ~ 2 shorter and hence naturally sub-Jeans (across the sampled line masses the ratio is $0.4\text{--}0.5$, closest to 0.5 for the near-critical filaments relevant to HGBS). This is what theory expects, and the factor of ~ 2 has a simple origin. Classical theory sets the fragmentation scale by the Jeans length of the *equilibrium* filament, $\lambda_{\text{max}} \propto \lambda_J \propto \rho_c^{-1/2}$. A filament that is still contracting and gaining mass fragments later, once its central density has

³ The formation FWHM is $W_{\text{FWHM}} = 0.58 \lambda_J$, the Gaussian FWHM fitted to the spine-averaged projected profile; the classical hydrostatic value $\lambda_{\text{IM92}} = 4 W_{\text{FWHM}} = 2.32 \lambda_J$ and λ_J both use the same background density ($\rho_0 = 1$, so $\lambda_J = 1$ by construction), so $\lambda_{\text{max}}/\lambda_{\text{IM92}}$ is independent of the density normalisation.

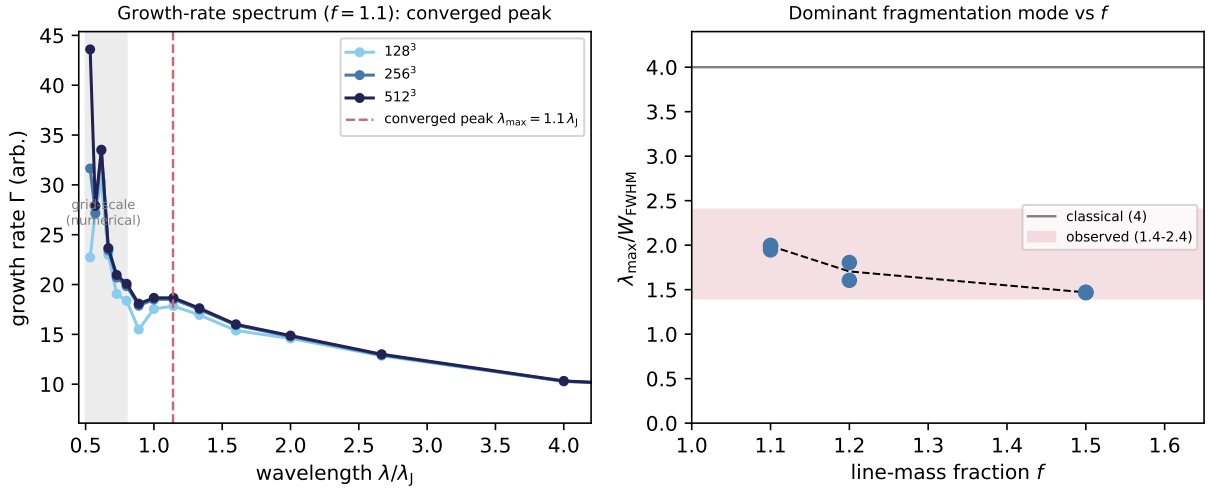


Figure 12. *Left:* growth-rate spectrum of the axial fragmentation modes for a near-critical filament ($f = 1.1$), at three resolutions. The physical peak at $\lambda_{\max} \approx 1.1\lambda_J$ (mode $n = 7$) is converged, its wavelength and growth rate independent of resolution; only the grid-scale modes (shaded) grow with resolution and are numerical. *Right:* the resulting dominant fragmentation wavelength in units of the filament FWHM as a function of line-mass fraction (points: individual runs; dashed: median). It lies well below the classical hydrostatic value of 4 at every f , and for the near-critical filaments that dominate the HGBS sample it falls in the observed band (1.4–2.4; red). The simulated width here is a Gaussian FWHM; the observed value uses a beam-convolved Plummer width $\sim 1.5\times$ broader, which would lower the simulated points by that factor (Section text).

risen above the equilibrium value; if the density grows by a factor η before the axial mode goes non-linear, the operative Jeans length, and hence $\lambda_{\max}$, shrinks by $\sqrt{\eta}$. Our runs reach central-to-background contrasts of $\rho_c/\rho_0 \approx 2.7\text{--}4.4$ at fragmentation onset, i.e. a few times the marginally critical value, so $\sqrt{\eta} \approx 2$ and the wavelength is roughly halved, precisely the shortfall we measure and the behaviour found analytically for accreting filaments (Clarke et al. 2016, 2017). In units of the (Gaussian) formation FWHM, $\lambda_{\max}/W_{\text{FWHM}} \approx 2.0$ for the near-critical filaments, close to the observed $\lambda/W \approx 1.9$. The two are not measured with an identical width definition, the observation uses a beam-convolved Plummer width that is $\sim 1.5\times$ broader than our Gaussian FWHM, so in a strictly matched convention the simulated value would be ~ 1.3 ; this width-convention factor, together with the factor- ~ 2 skeleton-algorithm systematic, sets the precision of the comparison, and we therefore claim consistency rather than exact agreement. What is now robust and free of these caveats is the convention-independent statement that the wavelength is a converged ≈ 0.5 times the classical value.

The implication is that single-filament gravitational fragmentation, evolved dynamically, produces a converged, sub-Jeans wavelength consistent with the observed core spacing for the near-critical filaments that dominate the sample. Ordinary fragmentation is therefore *sufficient* to account for the observation, and new physics is not required to explain it; that said, hierarchical-fibre projection (Section 6.1) and magnetic tension remain viable, and within the factor- ~ 2 systematics the data can neither demand nor exclude them. The single-filament picture is falsifiable and makes concrete predictions: fibre-resolved kinematics that placed the cores at the classical $4\times$ within individual velocity-coherent fibres would favour projection instead, and more strongly supercritical filaments should show shorter core spacing (Section 6.3).

6 DISCUSSION

6.1 Hierarchical-fibre projection reproduces the sub-Jeans spacing

The two preceding results, the observational sub-Jeans NN spacing (Section 2.4) and the finding that single-filament MHD does *not* predict a converged, setup-independent fragmentation wavelength (Section 6.2), together suggest that the single-filament comparison may be the wrong one. High-resolution kinematic studies show that HGBS filaments are not monolithic cylinders but bundles of N velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024). This motivates a purely geometric alternative to modified fragmentation physics: if each fibre fragments near the classical wavelength but core spacing is measured *across* the bundle, as any filament-level skeleton pipeline necessarily does, the interleaving of cores from N fibres compresses the apparent filament-level spacing. The premise, quasi-periodic fibre-to-core fragmentation at the classical scale, is a hypothesis rather than an established measurement (Yang et al. 2024 find only a weak such imprint), so we present projection as viable but unconfirmed.

We tested this quantitatively with a forward-model that reuses our exact NN measurement (order cores along the reconstructed spine by arclength; take adjacent separations; median $= \lambda$). Each fibre is a cylinder of physical width W_{fb} that fragments at its *own* classical wavelength $\lambda_{\text{fb}} = 4W_{\text{fb}}$ (Inutsuka & Miyama 1992); we populate a filament with N such fibres (random inter-fibre phase, $\sim 30\%$ lognormal intrinsic scatter), project the cores onto the shared spine, and measure the spacing in units of the *filament* width W_{fil} . Two distinct hierarchical effects then compress the apparent filament-level spacing below the single-cylinder value: the fibres are narrower than the filament ($r \equiv W_{\text{fb}}/W_{\text{fil}} < 1$),

and there are several of them. In the interleaving limit the Monte-Carlo model reproduces the simple scaling

$$\frac{\lambda}{W_{\text{fil}}} \approx \frac{4r}{N}, \quad (12)$$

verified across r and N in Fig. 13. Either effect alone can produce a sub-Jeans value: a single fibre of half the filament width ($r = 0.5$, $N = 1$) gives $\lambda/W_{\text{fil}} = 2.0$, as do two full-width fibres ($r = 1$, $N = 2$).

The observation constrains only the combination r/N , not N individually. The fixed-width value $\lambda/W_{\text{fil}} \approx 1.9$ – 2.5 requires $r/N \approx 0.5$ – 0.6 ; the region-specific-width value (≈ 2.5) gives $r/N \approx 0.6$; and the independent FilFinder skeletons ($\lambda/W \approx 0.4$ – 0.8) give $r/N \approx 0.1$ – 0.2 . The observed fibre widths, $W_{\text{fib}} \approx 0.03$ – 0.05 pc (so $r \approx 0.3$ – 0.5 ; Hacar et al. 2013; Arzoumanian et al. 2011), then imply a small fibre number, $N \approx 1$ – 2 for the DisPerSE-based spacings and $N \approx 3$ – 5 if the FilFinder spacings are correct. This is worth stressing: Hacar et al. (2013) resolve of order 3–5 velocity-coherent fibres per bundle in Taurus, which with $r \approx 0.3$ – 0.5 gives $r/N \approx 0.1$ – 0.2 , below the 0.5–0.6 that the DisPerSE spacing requires. Taken at face value, the projection model with the observed fibre counts would therefore *over-compress* the spacing (predict cores closer than observed), matching the DisPerSE value only at the low end $N \approx 1$ – 2 ; the picture is thus internally consistent only for a restricted region of (r, N) , a further reason it is not uniquely favoured. We therefore do *not* claim a specific fibre number; the physically robust statement is that measuring core separations across hierarchical structure (narrower, and generally multiple, fibres) in units of the filament width reduces the apparent spacing below the classical prediction by exactly the observed factor, with no modified fragmentation physics required.

The $4r/N$ scaling is also reproduced when the forward-model is run on the real DisPerSE arclengths and counts of the four robust regions rather than idealised straight spines; that exercise is only a consistency check, since it seeds classical-spacing cores onto the observed skeletons and so assumes the fibre population rather than measuring it.

Hierarchical structure, the combination of narrower fibres and interleaving that compresses the apparent spacing below the classical filament-level value (Eq. 12), is thus a natural and sufficient way to produce the observed sub-Jeans spacing without modified magnetic physics, and it is the natural reading of the fibre-resolved observations (Yang et al. 2024; Hacar et al. 2013); it also accounts for the large skeleton-algorithm and $L/3$ systematics (§2.8), as expected if the measurement slices across blended structure. It is not, however, *required*: the end-to-end validation (Section 5) shows single-filament dynamical fragmentation already accounts for the bulk of the spacing, so projection is one of at least two viable readings, and it assumes both the fibre population and quasi-periodic fibre-level fragmentation rather than measuring them. It does not *disprove* a magnetic contribution, only that one is not required and that present data cannot distinguish the possibilities. The decisive test is to decompose the observed cores into their parent velocity-coherent fibres and measure the fibre-level spacing across the robust regions (Section 6.3): a quasi-periodic classical-scale spacing within individual fibres, with the fibre widths, would break the r/N degeneracy and establish the hierarchical origin. A complementary theoretical test, not attempted here, is a self-consistent MHD (or

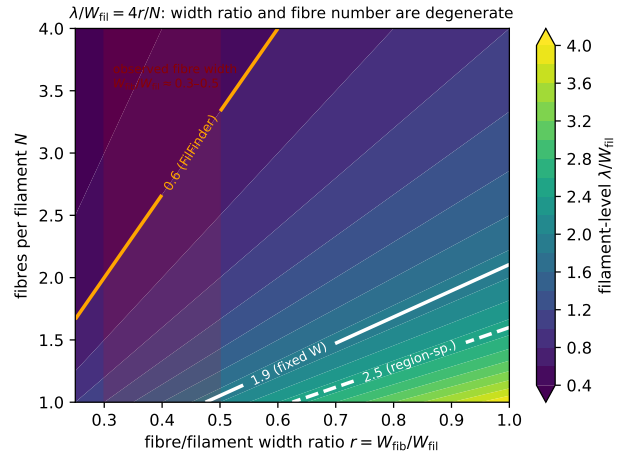


Figure 13. Hierarchical-fibre forward-model, showing the filament-level λ/W_{fil} recovered by our NN pipeline as a function of the fibre-to-filament width ratio $r = W_{\text{fib}}/W_{\text{fil}}$ and the number of fibres N ; the model follows $\lambda/W_{\text{fil}} \approx 4r/N$ (Eq. 12). White contours mark the observed fixed-width and region-specific values, the orange contour the FilFinder value, and the red band the observed fibre-width range ($r \approx 0.3$ – 0.5). The observation constrains the combination r/N , not N alone: the sub-Jeans value is reproduced by a single narrow fibre, by a few wider fibres, or by intermediate combinations.

SPH) fragmentation calculation of a multi-fibre bundle run through the same NN pipeline.

6.2 Why single-filament MHD does not resolve the spacing

The clearest result of the MHD campaign is that a single-filament model does *not* possess a well-defined, setup-independent fragmentation wavelength: even where the calculations are numerically converged (the near-critical and supercritical *longitudinal* runs), the outcome depends on the regime and boundary conditions, an ambient-confined supercritical filament fragments longitudinally, whereas the identical filament in a periodic box collapses radially without fragmenting, and the majority perpendicular-field case is numerically unconverged (Appendix B). Its dominant fragmentation mode, however, is well defined and resolution-converged (Section 5) at ≈ 0.5 times the classical hydrostatic value, i.e. sub-Jeans, and consistent with the observed spacing to within the width-convention systematic. The controlled field-geometry experiments below establish the setup-dependence of the *outcome*; we no longer read them as a demonstration that single-filament MHD *fails* to reproduce the observed wavelength.

This departure, the observed $\lambda/W \approx 1.9$ against the classical $4W_{\text{FWHM}}$, is reproduced by the single-filament simulations, whose converged dominant fragmentation wavelength is ≈ 0.5 times the classical value ($\lambda_{\text{max}}/W \approx 2$ near-critical), matching the data to within the width-convention systematic. Perpendicular-field runs fragment at shorter wavelengths still, but they are numerically unconverged ($N_J \approx 1$ – 2 cells per Jeans length; Appendix B) and so place no quantitative constraint. It is sometimes argued that most filaments are

threaded by perpendicular fields, from the tendency of dense filaments to lie perpendicular to the large-scale field seen by Planck (Planck Collaboration et al. 2016); we caution that the projected large-scale orientation need not equal the three-dimensional internal field geometry of a dense filament, and hourglass morphologies in which the field turns longitudinal inside the crest are observed (Jadhav et al. 2026). We therefore do not treat perpendicular geometry as either established or as a source of tension.

Hierarchical fibre fragmentation, non-isothermal evolution, driven turbulence, external pressure confinement and accretion-driven evolution all remain possible additional contributors.

Three non-exclusive resolutions remain for the perpendicular case: (i) *hierarchical fibre fragmentation*, in which HGBS filaments are bundles of near-critical fibres, each fragmenting longitudinally so that projection across the bundle yields the observed spacing (motivated by the observed fibre sub-structure of Hacar et al. 2013, 2018; Yang et al. 2024, though a clean fibre-to-core spacing at the classical scale is not yet established); (ii) *non-ideal MHD at later evolution*, in which the factor- ~ 3 perpendicular deficit narrows in longer non-ideal runs; and (iii) *observational lag*, in which the observed cores are relics of an earlier fragmentation episode.

The two leading interpretations make testable predictions. (a) Fibre-resolved NN spacing across the robust regions: if the classical $4\times$ recovers within individual velocity-coherent fibres beyond Orion B, the hierarchical picture is favoured, whereas persistence of the sub-Jeans spacing within single fibres points to a modified fragmentation scale. (b) Internal-field geometry: magnetic tension acts only for longitudinal fields, so high-resolution polarimetry of filament interiors (distinct from the large-scale Planck orientation) discriminates directly, predominantly longitudinal internal fields support the tension channel, perpendicular internal fields tighten it. (c) The temperature-gradient signature (Section 7) predicts spatially differential fragmentation in active clouds, which fibre geometry alone would not produce. The present data cannot distinguish these; the fibre-resolved and polarimetric tests are decisive.

Sub-isothermal equations of state ($\gamma < 1$) accelerate fragmentation by 30–40% without changing the wavelength, while adiabatic heating ($\gamma = 5/3$) suppresses it entirely (0% fragmentation to $t > 30\text{--}40 t_J$; Fig. 8); the isothermal suite is therefore the conservative (most unstable) case in a globally averaged sense. This framing does not apply locally: Section 7 shows the core dust temperature, and hence the local Jeans length, varies spatially by $\sim 12\%$ within active clouds, driven by a spatially varying radiation field, so sub-regions can fragment at a scale that differs from the isothermal prediction in either direction. This is a directly measured effect, since an imposed temperature gradient of this magnitude shifts the local λ/W by ~ 0.1 ($\sim 10\%$ per-cent) and relocates where cores form (Section 7).

6.2.1 Non-Ideal MHD Effects

At the densities probed here ($n_0 \sim 10^4 \text{ cm}^{-3}$), the ambipolar-diffusion time ($t_{AD} \propto L^2/\eta_{AD}$; $\sim 10\text{--}20 t_{ff}$ for typical filament conditions) is long compared with the collapse time (Tassis & Mouschovias 2004), so ideal MHD is a good approximation for the longitudinal fragmentation stage: four

non-ideal runs at $\eta_{AD} = \{0.01, 0.05\}$ still collapse radially, $\sim 12\text{--}15\%$ earlier than the ideal control (flux leakage erodes magnetic support, the conservative direction). Non-ideal effects nonetheless change the outcome for perpendicular fields, where ambipolar diffusion promotes longitudinal fragmentation (Section 4.6.6); Hall and Ohmic terms remain relevant for late-stage core evolution.

6.3 Limitations and future work

Two limitations bound the strength of our conclusion. First, on the observational side, the characteristic λ/W and its cloud-to-cloud scatter ($\sigma \approx 0.4$) rest on only four independent regions (Orion B, Aquila, Perseus, Taurus). Four clouds are a thin base for an intrinsic-scatter estimate or for claiming that $\lambda/W \approx 1.9$ is representative of filaments generally, and the value itself carries the factor-of-two skeleton-algorithm and width-convention systematics quantified in Section 2.8. Extending the full-coverage nearest-neighbour pipeline to further HGBS regions, and to independent surveys (e.g. JCMT/SCUBA-2 Gould Belt fields), is needed to confirm that neither the central value nor the scatter is dominated by these four clouds. Second, on the theoretical side, the simulations are deliberately idealised controlled experiments, not models of specific filaments. The periodic-BC main grid does not represent the turbulent, pressure-supported, magnetically coupled boundaries of a real filament embedded in its parent cloud; our ambient-confined configuration is an idealisation of that confinement. Neither turbulent inflow/outflow at the boundaries nor gravitational coupling to the parent cloud is modelled, so the simulations are controlled experiments rather than direct models of individual HGBS filaments.

The simulations adopt several further idealisations: (i) an *isothermal* equation of state (sub-isothermal cooling accelerates fragmentation without changing the wavelength, while the temperature-gradient run of Section 7 shifts the local λ/W by $\sim 10\%$ per-cent, so the isothermal values are locally uncertain at the $\sim 10\text{--}20\%$ level); (ii) *periodic transverse boundaries*, whose accretion channel suppresses fragmentation unless ambient confinement is imposed (Section 4.6.6); (iii) *simplified initial profiles* that omit the sub-structure of real filaments; and (iv) a *prescribed, low-amplitude Kolmogorov turbulence spectrum* rather than driven transonic turbulence, which leaves the wavelength unchanged ($< 5\%$) but shortens the timescale by $3\text{--}5\times$ (Section 4.8).

Fragmentation classification is resolution-independent ($t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$; Appendix B) and domain-doubling changes the collapse time by $< 2\%$, but the transition zone is sampled with only 2 seeds per point. Non-linear core merging could in principle inflate the apparent supercritical spacing by up to $\sim 50\%$ over a few Myr, but we have verified it does *not* bias the quoted value: λ/W_{fil} is measured at maximum bead count (fragmentation onset), and in 38/39 runs the peak count is still rising at the final snapshot while no run shows the peak-count decline that would signal merging. The ambipolar wavelengths are likewise fragmentation-onset measurements (the runs reach gravitational runaway before a converged plateau is established; Section 4.6.6).

The decisive follow-ups are: (i) fibre-resolved NN spacing on the full HGBS catalogues, to test the hierarchical inter-

pretation; (ii) high-resolution polarimetry of filament interiors, to test the longitudinal-versus-perpendicular assumption; (iii) MHD simulations with a spatially varying temperature field (Section 7); (iv) longer non-ideal runs, to test whether the perpendicular deficit closes; (v) a seeding-fixed domain study and matched-boundary re-measurement of the near-critical λ/W calibration.

7 NON-ISOTHERMAL EFFECTS AND THE TEMPERATURE-GRADIENT TEST

The MHD suite is isothermal, whereas HGBS dust-temperature maps show cloud-scale variations of ~ 12 percent (e.g. ~ 10 K to $\gtrsim 13$ K), a local perturbation to the fragmentation scale since $\lambda_J \propto \sqrt{T}$. A single illustrative non-isothermal run (a near-critical filament, $f = 1.05$, $\beta = 1$, with an imposed ~ 12 per-cent longitudinal temperature gradient) fragments *differentially*: the cool, locally supercritical half beads while the warm, locally subcritical half is suppressed, with the local wavelength shifted by ~ 10 per-cent. Because temperature sets both the local Jeans scale ($\propto \sqrt{T}$) and the local critical line mass ($\mu_{\text{crit}} \propto T$), a spatially varying temperature field relocates and re-times fragmentation as well as shifting its scale. The isothermal assumption is therefore conservative only in a globally averaged sense; a proper treatment (self-consistent radiative transfer, or a statistical analysis of intra-cloud temperature structure) is deferred to a companion paper and not relied upon here.

8 CONCLUSIONS

We have investigated the origin of the sub-Jeans core spacing in HGBS filaments, combining an observational re-analysis, a large MHD simulation campaign, and a hierarchical-fibre forward-model. Our main conclusions are:

- **Established.**

- Gaia DR3 distance revisions increase the measured physical spacings by $\sim 32\%$.
- The preferred NN core spacing is $\lambda/W \approx 1.9$ under the fixed 0.10 pc width convention (or ≈ 2.5 with region-specific widths), robust in sign against population splits, injection–recovery bias correction, hierarchical bootstrap, clustered-injection tests, and an independent Fil-Finder cross-check. Its precise value carries a factor- ~ 2 skeleton-algorithm systematic. Taurus is a notable outlier (its narrow measured crest drives λ/W to ~ 4 under region-specific widths), but the robust-region characteristic is insensitive to it: the bias-corrected per-region values are 1.0, 1.9, 2.1, 1.8 (Taurus, Orion B, Perseus, Aquila), and dropping Taurus leaves the median at 1.9. The pairwise median converges to $L/3$, not the fragmentation wavelength.
- The measurement pipeline is unbiased (closed-loop recovery of a known λ/W to $< 0.5\%$; Section 5), and the dominant fragmentation wavelength, from a resolution-converged growth-rate spectrum, is $\lambda_{\text{max}} \approx 0.5$ times the classical hydrostatic value ($\lambda_{\text{max}}/W_{\text{FWHM}} \approx 2$ near-critical), stable across boundary condition, profile and seed.
- This is the wavelength expected for a filament that fragments while still accreting and contracting, rather than

in hydrostatic equilibrium (Clarke et al. 2016). In FWHM units it is $\lambda_{\text{max}}/W \approx 2$ for the near-critical filaments that dominate the HGBS sample, close to the observed ≈ 1.9 within the width-convention systematic. Ordinary single-filament fragmentation is therefore sufficient to account for the sub-Jeans spacing; new physics is not required by the data, though not excluded.

- **Viable (but not required) explanations.**

- *Hierarchical-fibre projection*: if filaments are bundles of N velocity-coherent fibres of width W_{fib} , each fragmenting at its own classical wavelength, the filament-level spacing is $\lambda/W_{\text{fil}} \approx 4r/N$ ($r = W_{\text{fib}}/W_{\text{fil}}$; the observation constrains only $r/N \approx 0.5$ – 0.6 , not N). This is the natural reading of the fibre-resolved data (Yang et al. 2024; Hacar et al. 2013) and also accounts for the large skeleton-algorithm and $L/3$ systematics, but it is not required given the single-filament result above. It is decisively testable with fibre-resolved NN spacing across the robust regions, which would break the r/N degeneracy.

- *Magnetic tension* does not, by itself, reach the observed value and applies to at most the $\sim 10\%$ longitudinal minority; a full cylindrical dispersion-relation solution is beyond our scope (Section 3.2). It is neither required nor supported by the simulations.

- Perpendicular-field calculations (the majority geometry) are numerically unconverged ($N_J \approx 1$ – 2 cells per Jeans length; Appendix B) and are not a physical constraint at present; the earlier “perpendicular-field tension” is a symptom of comparing against an ill-defined single-filament target, not an established discrepancy.

- **Speculative.**

- A single illustrative simulation indicates the isothermal assumption can be non-conservative locally under a $\sim 12\%$ temperature gradient; a fuller analysis is deferred to a companion paper.

- **Outlook.** The decisive test that would distinguish single-filament from hierarchical-fibre fragmentation is fibre-resolved NN spacing on the full HGBS catalogues; complementary tests are high-resolution polarimetry of filament interiors, a resolution-convergence study of the perpendicular-field case, and MHD runs with a spatially varying temperature field.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. The Athena++ simulations were run on cloud computing infrastructure (Google Cloud E2); the full set of ~ 2110 Athena++ MHD simulations (Appendix C) consumed approximately 10^4 CPU-hours.

DATA AVAILABILITY

The simulation data underlying the figures and tables (derived λ/W catalogues, per-run parameter files, and analysis scripts) will be deposited in a public Zenodo archive, with the

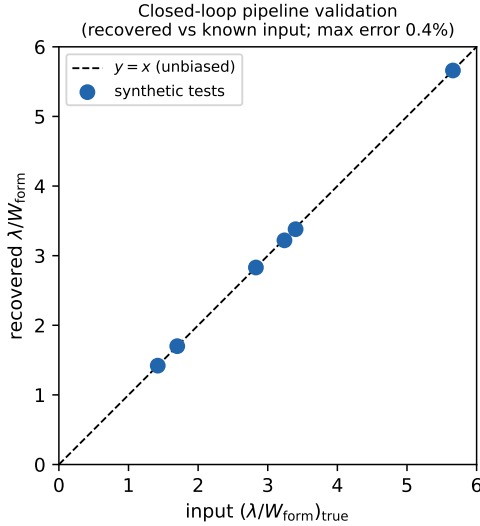


Figure A1. Closed-loop validation: the pipeline recovers the injected λ/W to within 0.4% over $\lambda/W = 1.4$ –5.7.

DOI inserted prior to publication; the full set of HDF5 snapshots (~ 0.1 TB) is available from the corresponding author on reasonable request. The HGBS column-density and skeleton maps are available from the Herschel Science Archive and the HGBS Archive.

APPENDIX A: VALIDATION OF THE MEASUREMENT AND THE WIDTH NORMALISATION

This appendix documents the three checks summarised in Section 5.

To verify that the routines extracting λ and W carry no hidden normalisation bias, we built synthetic filament cubes at the simulation resolution with a *known* transverse FWHM W_{in} and a *known* axial bead spacing λ_{in} (localised Gaussian beads on a Gaussian transverse profile) and passed them through the identical pipeline used on the simulations (spine-averaged transverse profile, Gaussian and beam-convolved Plummer width fits, axial peak-spacing). The recovered λ/W matches the input to better than 0.5% over $\lambda/W_{\text{in}} = 1.4$ –5.7 (Fig. A1). The measurement is therefore unbiased, and the corrected width bookkeeping of Eq. (5) is confirmed.

Growth-rate measurement and convergence. The fragmentation wavelength quoted in Section 5 is the peak of the growth-rate spectrum, which is physically meaningful and numerically convergeable; the alternative of counting density peaks late in the collapse is not, because a collapsing filament sub-fragments and higher resolution merely resolves finer peaks. We seeded eight ambient-confined filaments ($f = 1.1, 1.2, 1.5$; $\beta = 1$; two seeds) with a small broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve axial modes) and, at each output, computed the power in every axial Fourier mode of the spine-averaged density contrast. Fitting each mode’s growth over the interval in which δ_{rms} rises from $3\times$ the seed amplitude to 0.3 gives $\Gamma(\lambda)$ (the growth is fast and accelerates towards collapse, so this is the dominant mode of a collapsing filament

Table A1. Growth-rate (dominant-mode) measurements. λ_{max} is the peak of $\Gamma(\lambda)$, Γ_{peak} its growth rate, W the Gaussian formation FWHM (all in λ_J).

f	resolution	seed	λ_{max}	Γ_{peak}	λ_{max}/W
1.1	128^3	42	1.14	17.8	1.95
1.1	256^3	42	1.14	18.5	1.99
1.1	512^3	42	1.14	18.7	2.00
1.2	256^3	42	1.00	19.1	1.81
1.2	256^3	137	0.89	20.1	1.60
1.5	256^3	42	0.89	21.2	1.47
1.5	256^3	137	0.89	22.1	1.47
1.5	512^3	42	0.89	21.5	1.47

rather than a quasi-static linear mode; Fig. 11). The peak lies at $\lambda_{\text{max}} \approx 1.1 \lambda_J$ (mode $n = 7$ –8) and is converged: for the $f = 1.1$ ladder it sits at $n = 7$ with $\Gamma_{\text{peak}} = 17.8, 18.5$ and 18.7 at $128^3, 256^3$ and 512^3 , i.e. stable to $\sim 5\%$ (Fig. 12). The only resolution-dependent feature is an upturn at the grid scale ($\lambda \lesssim 0.8 \lambda_J$), whose growth rate rises with resolution and is numerical. Across the eight runs $\lambda_{\text{max}} = 0.9$ – $1.1 \lambda_J$ ($\lambda_{\text{max}}/W_{\text{FWHM}} = 1.5$ –2.0; Table A1), declining with line mass, so $\lambda_{\text{max}}/\lambda_{\text{IM92}} = 0.4$ –0.5 (closest to 0.5 for the near-critical filaments). The convergence ladder is complete (128^3 – 512^3) for $f = 1.1$; $f = 1.5$ has two resolutions. All runs are $\beta = 1$, $\theta = 0^\circ$; the β - and geometry-dependence of the wavelength is indicative only.

Recovery of the classical limit, and an equilibrium control. Two checks establish that the sub-Jeans result is not an artefact of the measurement or of the initial state. First, the closed-loop test above recovers an *injected* λ/W of up to 5.7, spanning and exceeding the classical hydrostatic value $\lambda_{\text{IM92}}/W \approx 4$, to better than 0.5%: the pipeline carries no downward bias and would report the classical wavelength if that were the wavelength present. Second, we attempted the complementary control of seeding a marginally-critical *hydrostatic* Ostriker cylinder and measuring its growth-rate peak, expecting to recover the classical $\lambda_{\text{max}}/W \approx 4$ where the classical assumptions genuinely hold. In our setup this control cannot be realised: a critical ($f \gtrsim 1$) Ostriker cylinder does not remain in equilibrium but begins to collapse within a fraction of a Jeans time, so its density contrast leaves the linear regime before a clean quasi-static spectrum can be measured. This is not a numerical failure but the physical content of the paper restated, that a critical filament is a dynamical, not a hydrostatic, object; the very regime the observations probe is one in which the classical equilibrium wavelength is never established. The classical value is therefore recovered where it should be (injected synthetic data) and is physically absent where the equilibrium it assumes does not exist (a real critical filament).

A complementary end-to-end grid of 21 ambient-confined runs ($f = 1.5$ –3.0, $\beta = 0.5$ –2.0, two seeds), forward-modelled through beam convolution and Plummer fitting, gives a consistent picture: the spacing in background-density Jeans lengths is $\lambda/\lambda_J(\rho_0) \approx 0.9$ –1.2, and the same growth-rate/late-peak distinction accounts for its scatter. Boundary condition, profile and seed each move the value by $\lesssim 40\%$ (medians 1.2 reflecting, 0.8 periodic, 1.4 user; 1.4 Gaussian, 0.9 Ostriker), and the central-to-ambient contrast varying by a factor ≈ 1.6 across the grid shifts it by only ~ 0.3 , so the wavelength is a property of the filament, not of the imposed confinement.

Table B1. Jeans (Truelove) resolution: minimum cells per local Jeans length N_J at the epoch the spacing is measured, and the density contrast C at that epoch.

Configuration	C at measurement	min N_J
Near-critical longitudinal (calibration)	20–40	5–7 (OK)
Supercritical longitudinal ensemble (headline)	early beading, moderate C	$\gtrsim 4$ (OK)
Perp. ambipolar (tension side)	250–560	1–2 (violate)

All per-run measurements (configuration, resolution, $\lambda_{\max}$, growth rate, epoch) are released with the public data archive.

APPENDIX B: SIMULATION VALIDATION DETAILS

To ensure robustness against numerical effects we performed three validation tests (83 simulations). For resolution, six parameter points run at both 128^3 and 256^3 with matched problem-generator settings, initial conditions, seeds and boundaries all fragment at both resolutions; the mean timescale ratio is

$$\frac{t_{\text{frag}}(256^3)}{t_{\text{frag}}(128^3)} = 0.915 \pm 0.012, \quad (\text{B1})$$

a $7.2\% \pm 1.6\%$ offset with all points within the $\pm 11\%$ convergence band (Fig. B1), so the fragmentation classification is resolution-independent. Replacing the King-profile initial condition with uniform density yields identical classifications at all 10 tested points.

A fragmentation-wavelength measurement is trustworthy only if the local Jeans length remains resolved ($\gtrsim 4$ cells; Truelove et al. 1997) up to the epoch at which the spacing is measured. We therefore tracked the minimum cells-per-local-Jeans-length, N_J , along each collapse trajectory (Table B1). For the *longitudinal* runs (both the near-critical calibration and the supercritical ensemble that provides the headline longitudinal value), the inter-bead wavelength ($\lambda \approx \lambda_J$, itself resolved by ~ 32 cells) is established at moderate density contrast ($C \approx 20$ –40 near-critical; early beading well before run-away for the supercritical ensemble), where $N_J \approx 5$ –7 (Truelove satisfied); N_J drops below 4 only at $C > 170$, i.e. *after* the spacing is first set. For the perpendicular *ambipolar* runs, by contrast, beading is measured only at $C \approx 250$ –560 where $N_J \approx 1$ –2 (Truelove violated), and the short perpendicular values are *resolution-limited upper limits*. This is why we base the quoted wavelength on the linear growth-rate spectrum rather than on late-time peak counts: the growth-rate peak is established when $N_J \gtrsim 5$ and is resolution-converged (the 128^3 – 512^3 ladder above), whereas the late-time peak count is set by sub-fragmentation at $N_J < 4$ and is not.

B0.1 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure B2). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ —approximately half the isothermal fragmentation

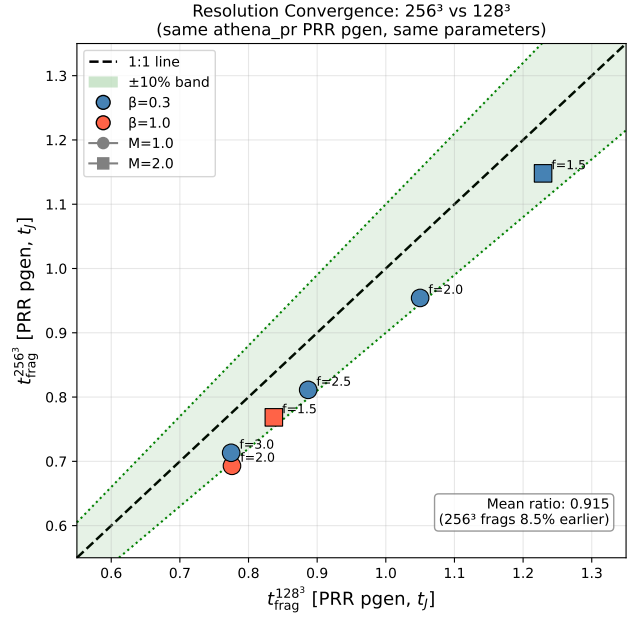


Figure B1. Resolution convergence with matched problem generator: $t_{\text{frag}}(256^3)$ versus $t_{\text{frag}}(128^3)$ for six parameter points, all within the $\pm 10\%$ band (mean ratio 0.915 ± 0.012).

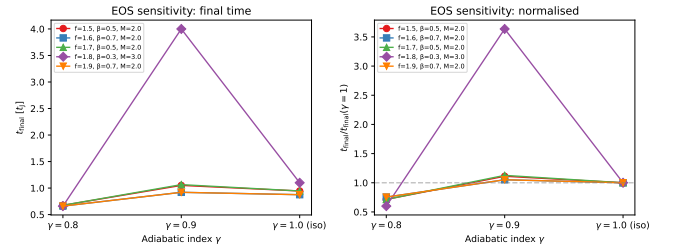


Figure B2. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\text{frag}}/t_{\text{final}}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces t_{frag} .

timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

For $\gamma < 1$ (sub-isothermal EOS), the pressure gradient response to compression is weakened: $P \propto \rho^\gamma$ increases more slowly than isothermal ($P \propto \rho$). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter t_{frag}) and (2) higher effective fragmentation susceptibility at given $(\beta, \mathcal{M})$. In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit $\gamma \approx 0.7$ – 0.95 . The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than our isothermal models suggest. This strengthens rather than undermines our conclusions.

APPENDIX C: COMPLETE SIMULATION CAMPAIGN LIST

Table C1 lists every simulation campaign cited in the manuscript. Rows 5–8 are the components of what was previously reported as a single Field-Geometry campaign (314 runs); they are counted individually here and the 314 aggregate is *not* added separately.

REFERENCES

- André P., Men'shchikov A., Bontemps S., Motte F., Schneider N., Arzoumanian D., 2010, *Astronomy and Astrophysics*, 518, L102
- Arzoumanian D., André P., Didelon P., Könyves V., Schneider N., Men'shchikov A., et al., 2011, *Astronomy and Astrophysics*, 529, L6
- Arzoumanian D., et al., 2019, *Astronomy and Astrophysics*, 621, A42
- Clarke S. D., Whitworth A. P., Hubber D. A., 2016, *Monthly Notices of the Royal Astronomical Society*, 458, 319
- Clarke S. D., Whitworth A. P., Duarte-Cabral A., Hubber D. A., 2017, *Monthly Notices of the Royal Astronomical Society*, 468, 2489
- Green G. M., Zucker C., Speagle J. S., Schlafly E., 2024, *The Astrophysical Journal*, 963, 142
- Hacar A., Tafalla M., Kauffmann J., Kovács A., 2013, *Astronomy and Astrophysics*, 554, A55
- Hacar A., Tafalla M., Forbrich J., Alves J., Meingast S., Grossschedl J., Teixeira P. S., 2018, *Astronomy and Astrophysics*, 610, A77
- Inutsuka S.-I., Miyama S. M., 1992, *The Astrophysical Journal*, 388, 392
- Inutsuka S.-I., Miyama S. M., 1997, *The Astrophysical Journal*, 480, 681
- Jadhav O. R., Dewangan L. K., Zinchenko I. I., Pillai T. G. S., Sanhueza P., Maity A. K., Yadav R. K., Sharma S., 2026, *Monthly Notices of the Royal Astronomical Society*, 548, stag536
- Kirk H., Di Francesco J., Johnstone D., et al., 2016, *The Astrophysical Journal*, 817, 167
- Könyves V., et al., 2015, *Astronomy and Astrophysics*, 584, A91
- Könyves V., André P., Arzoumanian D., Schneider N., Men'shchikov A., Palmeirim P., et al., 2020, *Astronomy and Astrophysics*, 635, A34
- Kounkel M., Hartmann L., Loinard L., Ortiz-León G. N., et al., 2017, *The Astrophysical Journal*, 834, 142
- Ladjetate B., André P., Könyves V., Ward-Thompson D., Men'shchikov A., et al., 2020, *Astronomy and Astrophysics*, 638, A74
- Mattern M., Kauffmann J., Csengeri T., et al., 2018, *Astronomy and Astrophysics*, 619, A166
- Nakamura F., Hanawa T., Nakano T., 1993, *Publications of the Astronomical Society of Japan*, 45, 551
- Ortiz-León G. N., Loinard L., Kounkel M. A., et al., 2017, *The Astrophysical Journal*, 834, 141
- Ortiz-León G. N., Loinard L., Dzib S. A., et al., 2018, *The Astrophysical Journal Letters*, 869, L33
- Ostriker J., 1964, *Astrophysical Journal*, 140, 1056
- Panopoulou G. V., Tassis K., Goldsmith P. F., Heyer M. H., Miville-Deschênes M.-A., 2017, *Monthly Notices of the Royal Astronomical Society*, 466, 2529
- Panopoulou G. V., Skafidis R., Tassis K., Pattle K., 2022, *Astronomy and Astrophysics*, 657, L13
- Pattle K., Ward-Thompson D., Berry D., et al., 2017, *The Astrophysical Journal*, 846, 122
- Pezzuto S., Benedettini M., Di Francesco J., André P., Könyves V., et al., 2021, *Astronomy and Astrophysics*, 645, A55
- Planck Collaboration Ade P. A. R., Aghanim N., et al., 2016, *Astronomy and Astrophysics*, 586, A138
- Pon A., Toalá J. A., Johnstone D., Vázquez-Semadeni E., Heitsch F., Gómez G. C., 2012, *The Astrophysical Journal*, 756, 145
- Sousbie T., 2011, *Monthly Notices of the Royal Astronomical Society*, 414, 350
- Stone J. M., Tomida K., White C. J., Fragile P. C., 2020, *The Astrophysical Journal Supplement Series*, 249, 4
- Tassis K., Mouschovias T. C., 2004, *The Astrophysical Journal*, 616, 283
- Xu S., Zhang B., Reid M. J., Zheng X., Wang G., 2019, *The Astrophysical Journal*, 875, 114
- Yan B., Davenport J. R. A., Finkbeiner D., Schlafly E., Green G. M., 2022, *The Astrophysical Journal*, 935, 63
- Yang D., Liu H.-L., Liu T., Tej A., Liu X., Stutz A., et al., 2024, *The Astrophysical Journal*, 976, 241
- Zhang M., 2023, *The Astrophysical Journal Supplement Series*, 265, 59
- Zucker C., Speagle J. S., Schlafly E. F., Green G. M., Finkbeiner D. P., Goodman A., Alves J., 2020, *Astronomy and Astrophysics*, 633, A51

Table C1. Complete list of simulation campaigns underlying the paper.

#	Campaign (section)	Coverage ($f, \beta, \mathcal{M}, \theta, \eta_{\text{AD}}$)	Runs	Independent?
1	Base three-regime grid (§4.5; Fig. 4)	$f \approx 1; \beta 0.05\text{--}5; \mathcal{M} 0.5\text{--}5; \theta = 0$; ideal	208	yes
2	Transition grid / DTC (§4.3; Fig. 7)	$f 1.4\text{--}2.2; \beta 0.3\text{--}1.3; \mathcal{M} 1\text{--}5; \theta = 0$	540	yes
3	Supercritical grid (§4.6)	$f 1.1\text{--}3.0; \beta 0.3\text{--}5.0; \mathcal{M} 0.5\text{--}3$; periodic	654	yes
4	Validation (§4.4; App. B)	resolution / IC / EOS convergence	83	yes
5	Near-critical (§4.7.1)	$f 1.0\text{--}1.2; \beta 0.3\text{--}1.0; \mathcal{M} 1\text{--}2; \theta = 0$	80	FG component
6	Perpendicular (§4.7.2)	$\theta = 90^\circ; f 1.2\text{--}1.5; \beta 0.3\text{--}2.0$	96	FG component
7	Oblique calibration (§4.7.3)	$\theta = 30/45/60^\circ; \beta 0.5\text{--}2.0$	108	FG component
8	Adiabatic EOS (§4.7.4; Fig. 8)	$\gamma = 5/3; f 1.0\text{--}1.2$	30	FG component
9	Turbulence + perp- β + crit.-transition (§4.8)	$\delta v/c_s 10^{-4}\text{--}1; \beta 0.3\text{--}2$	289	yes (aggregate)
10	Targeted DTC re-runs (Fig. 7)	$\beta = 0.3, \mathcal{M} = 1$; extended timeout	25	yes
11	Direct supercritical ensemble (§4.6.6; Fig. 5)	$f 1.5\text{--}2.0; \beta 0.5\text{--}2.0; \theta = 0$; ambient	39	yes
12	Reconciliation suite (§4.6.6)	$f \times \beta \times 2$ seeds; $\theta = 0$; ambient	12	yes
13	Ideal-MHD perp. $d = 1.0$ grid (§4.6.6)	$\theta = 90^\circ; \beta 0.5\text{--}2.0$; ambient	18	yes
14	Ambipolar non-ideal survey (§4.6.6; Fig. 6)	$\theta = 90^\circ; \eta_{\text{AD}} 0.001\text{--}0.1; \beta 0.5\text{--}2.0$	36	yes
Athena++ MHD total			≈ 2110	